



Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

- VERIFIED 0.80 - 1.00** 3+ independent sources including media
- CONFIRMED 0.50 - 0.79** 2 sources or 1 high-quality media
- EMERGING 0.25 - 0.49** Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	266	Media
TechCrunch AI	6	Media
HackerNews	4	Community
MIT Tech Review	3	Media
The Verge AI	3	Media

VERIFIED INTELLIGENCE — 4 stories

VERI

All of the updates from Elon Musk and Sam Altman's battle over OpenAI

The Verge AI

+2 sources

100%

Sam Altman and Elon Musk are facing off in a high-stakes trial that could alter the future of OpenAI and its most well-known product, ChatGPT. In 2024, Musk filed a lawsuit accusing OpenAI of abandoning its founding mission of developing AI to benefit humanity and shifting focus...

<https://www.theverge.com/tech/917225/sam-altman-elon-musk-openai-lawsuit>

VERI

Counterparty Modeling is Not Strategy: The Limits of LLM Negotiators

ArXiv CS.AI

+1 source

80%

arXiv:2605.16575v1 Announce Type: new Abstract: Negotiation requires more than inferring what the other side wants: it requires using that information to make advantageous offers and counteroffers over multiple turns. We study whether large language model (LLM) agents do this in...

<https://arxiv.org/abs/2605.16575>

VERI

Extending Pretrained 10-Second ECG Foundation Models to Longer Horizons

ArXiv CS.AI

+1 source

80%

arXiv:2605.16975v1 Announce Type: cross Abstract: Electrocardiogram (ECG) foundation models pretrained on typical diagnostic 10-second ECG segments, have demonstrated strong transferability across a range of clinical applications. However, many real-world applications produce...

<https://arxiv.org/abs/2605.16975>

VERI

Responsible Agentic AI Requires Explicit Provenance

ArXiv CS.AI

+1 source

80%

arXiv:2605.17169v1 Announce Type: new Abstract: Agentic AI is rapidly proliferating across diverse real-world domains such as software engineering, yet public trust has not kept pace. The central reason is that responsibility, despite being widely discussed, remains a subjective...

<https://arxiv.org/abs/2605.17169>

CONFIRMED SIGNALS — 273 stories

CONF

Show HN: Sx - an open-source package manager for AI skills, MCPs, and commands

HackerNews

50%

HN 50

No summary — see link for full article.

<https://github.com/sleuth-io/sx>

CONF

Show HN: Beacon - The open-source layer for local AI agent visibility

HackerNews

50%

HN 19

I built IONA over 10 years, working alone. It's a complete sovereign digital ecosystem — OS (PC + Phone), L1 blockchain, programming language, and on-device AI — all written from scratch in Rust, without Linux, BSD, or Windows code. What's in the GitHub org...

<https://github.com/Asymptote-Labs/agent-beacon>

CONF

Show HN: Vecdb - local-first hybrid vector database in Rust (HNSW and BM25)

HackerNews

50%

HN 4

No summary — see link for full article.

<https://github.com/zaydmulani09/vecdb>

CONF

Show HN: Tracecast - open-source generative data apps built on top of Marimo

HackerNews 50% HN 3

Hi HN, I'm Malachy, the founder of Tracecast. This project lets you generate interactive data apps on top of your data, using a Cursor-style AI chat. It stitches together Marimo, LangGraph agents, and data warehouse query tools. It has an Apache 2.0 license. The initial use case...

https://github.com/tracecast/open_data_apps

CONF

SandboxAQ brings its drug discovery models to Claude — no PhD in computing required

TechCrunch AI 50%

Other venture-backed companies like Chai Discovery and Isomorphic Labs have raced to build better models. SandboxAQ is betting that access is the bigger obstacle and that Claude solves it.

<https://techcrunch.com/2026/05/18/sandboxaq-brings-its-drug-discovery-models-to-claude-...>

0 CONF

Anthropic has acquired the dev tools startup used by OpenAI, Google, and Cloudflare

TechCrunch AI 50%

Stainless, a New York-based startup, founded in 2022, rose to prominence in the emerging AI industry for automating the creation and maintenance of software development kits, or SDKs — the libraries developers use to interact with APIs.

<https://techcrunch.com/2026/05/18/anthropic-has-acquired-the-dev-tools-startup-used-by-...>

1 CONF

South Korea's LetinAR is building optics behind AI glasses

TechCrunch AI 50%

A lens the size of a thumbnail — and the South Korean startup that makes it — could become the optical backbone of the AI glasses era.

<https://techcrunch.com/2026/05/18/south-koreas-letin-ar-is-building-the-optics-behind-ai-...>

2 CONF

The Download: China's AI drama factory and the WHO's missing health targets

MIT Tech Review 50%

This is today's edition of The Download, our weekday newsletter that provides a daily dose of what's going on in the world of technology. How Chinese short dramas became AI content machines China's short drama industry is fueled by bite-sized, melodramatic, and smutty shows...

<https://www.technologyreview.com/2026/05/15/1137341/the-download-china-short-drama-ai-w-...>

3 CONF

Musk v. Altman proved that AI is led by the wrong people

The Verge AI 50%

The tech trial of the year, Musk v. Altman, was ultimately a fight for control. Elon Musk argued that Sam Altman, with whom he helped found the now-massive company OpenAI, shouldn't direct the future of AI. Altman's lawyers, in turn, poked at Musk's own credibility. A jury came...

<https://www.theverge.com/ai-artificial-intelligence/932464/musk-v-altman-proved-that-ai-...>

4 CONF

Amazon Alexa Plus can now create AI-generated podcasts

The Verge AI 50%

Alexa Plus, Amazon's upgraded AI assistant, can now generate podcasts on "virtually any topic," according to an announcement on Monday. With the update, Amazon says you can give Alexa Plus a topic, and the AI assistant will offer an overview of what its AI hosts plan to talk...

<https://www.theverge.com/tech/932375/amazon-alexa-plus-ai-podcasts>

5 CONF

TOBench: A Task-Oriented Omni-Modal Benchmark for Real-World Tool-Using Agents

ArXiv CS.AI 50%

arXiv:2605.16909v1 Announce Type: new Abstract: Tool-using agents are increasingly expected to operate across realistic professional workflows, where they must interpret multimodal inputs, coordinate external tools, inspect intermediate artifacts, and revise their actions before...

<https://arxiv.org/abs/2605.16909>

6 CONF

Harnessing LLM Agents with Skill Programs

ArXiv CS.AI 50%

arXiv:2605.17734v1 Announce Type: new Abstract: Equipping LLM agents with reusable skills derived from past experience has become a popular and successful approach for tackling complex and long-horizon tasks. However, such lessons are often encoded as textual guidance that...

<https://arxiv.org/abs/2605.17734>

7 CONF

RadGame: An AI-Powered Platform for Radiology Education

ArXiv CS.AI 50%

arXiv:2509.13270v2 Announce Type: replace-cross Abstract: We introduce RadGame, an AI-powered gamified platform for radiology education that targets two core skills: localizing findings and generating reports. Traditional radiology training is based on passive exposure to cases...

<https://arxiv.org/abs/2509.13270>

8 CONF

GCE-MIL: Faithful and Recoverable Evidence for Multiple Instance Learning in Whole-Slide Imaging

ArXiv CS.AI 50%

arXiv:2605.17456v1 Announce Type: cross Abstract: Multiple instance learning (MIL) is the standard approach for whole-slide image (WSI) classification and survival prediction, where attention-based models aggregate patch features into slide-level predictions. These models treat...

<https://arxiv.org/abs/2605.17456>

9 CONF

TTE-Flash: Accelerating Reasoning-based Multimodal Representations via Think-Then-Embed Tokens

ArXiv CS.AI 50%

arXiv:2605.16638v1 Announce Type: new Abstract: Recent research has demonstrated that Universal Multimodal Embedding (UME) benefits significantly from Chain-of-Thought (CoT) reasoning. In this paradigm, a generative model produces explicit reasoning traces for a multimodal...

<https://arxiv.org/abs/2605.16638>

0 CONF

Sustainable Intelligence for the Wild: Democratizing Ecological Monitoring via Knowledge-Adaptive Edge Expert Agents

ArXiv CS.AI 50%

arXiv:2605.16671v1 Announce Type: new Abstract: Rapid biodiversity loss underscore the urgency of effective monitoring, yet manual surveys remain resource-intensive. While on-device AI offers a scalable alternative, its performance in the wild is often challenged by...

<https://arxiv.org/abs/2605.16671>

1 CONF

SkillGenBench: Benchmarking Skill Generation Pipelines for LLM Agents

ArXiv CS.AI 50%

arXiv:2605.18693v1 Announce Type: new Abstract: As LLM agents are increasingly built around reusable skills, a central challenge is no longer only whether agents can use provided skills, but whether they can generate correct, reusable, and executable skills from repositories and...

<https://arxiv.org/abs/2605.18693>

2 CONF

Enhancing Metacognitive AI: Knowledge-Graph Population with Graph-Theoretic LLM Enrichment

ArXiv CS.AI 50%

arXiv:2605.16676v1 Announce Type: new Abstract: Metacognition-the ability to monitor one's own knowledge state, spot gaps, and autonomously fill them--remains largely absent from modern AI. Here, we present MetaKGenrich, a fully automated pipeline that endows large language...

<https://arxiv.org/abs/2605.16676>

3 CONF

State Contamination in Memory-Augmented LLM Agents

ArXiv CS.AI 50%

arXiv:2605.16746v1 Announce Type: new Abstract: LLM agents increasingly rely on persistent state, including transcripts, summaries, retrieved context, and memory buffers, to support long-horizon interaction. This makes safety depend not only on individual model outputs, but also...

<https://arxiv.org/abs/2605.16746>

4 CONF

Modelling Customer Trajectories with Reinforcement Learning for Practical Retail Insights

ArXiv CS.AI 50%

arXiv:2605.18449v1 Announce Type: cross Abstract: Understanding customer movement within retail spaces is essential for optimizing store layouts. Real-world trajectory data can provide highly accurate insights, but collecting it is costly and often infeasible for many retailers....

<https://arxiv.org/abs/2605.18449>

5 CONF

Voices in the Loop: Mapping Participatory AI

ArXiv CS.AI 50%

arXiv:2605.16827v1 Announce Type: new Abstract: Participatory approaches to artificial intelligence are increasingly documented across public, civic, and humanitarian settings, but evidence about how participation is organized remains fragmented. This paper reports on the...

<https://arxiv.org/abs/2605.16827>

6 CONF

CyberCorrect: A Cybernetic Framework for Closed-Loop Self-Correction in Large Language Models

ArXiv CS.AI 50%

arXiv:2605.17305v1 Announce Type: new Abstract: Large language model (LLM) self-correction -- the ability to detect and fix errors in generated outputs -- remains largely ad hoc, relying on generic prompts such as "please reconsider your answer" without systematic error analysis...

<https://arxiv.org/abs/2605.17305>

7 CONF

Vision-OPD: Learning to See Fine Details for Multimodal LLMs via On-Policy Self-Distillation

ArXiv CS.AI 50%

arXiv:2605.18740v1 Announce Type: cross Abstract: Multimodal Large Language Models (MLLMs) still struggle with fine-grained visual understanding, where answers often depend on small but decisive evidence in the full image. We observe a regional-to-global perception gap: the same...

<https://arxiv.org/abs/2605.18740>

8 CONF

Virtual Nodes Guided Dynamic Graph Neural Network for Brain Tumor Segmentation with Missing Modalities

ArXiv CS.AI 50%

arXiv:2605.16880v1 Announce Type: new Abstract: Multimodal magnetic resonance imaging (MRI) is crucial for brain tumor segmentation, with many methods leveraging its four key modalities to capture complementary information for effective sub-region analysis. However, the absence...

<https://arxiv.org/abs/2605.16880>

9 CONF

A Theory of Training Profit-Optimal LLMs

ArXiv CS.AI 50%

arXiv:2605.16430v1 Announce Type: cross Abstract: Scaling LLMs requires tremendous computational resources, and recent advances in AI have gone hand in hand with massive amounts of capital expenditure. While it is established that scaling up LLMs reliably increases model quality...

<https://arxiv.org/abs/2605.16430>

0 CONF

How do Humans Process AI-generated Hallucination Contents: a Neuroimaging Study

ArXiv CS.AI 50%

arXiv:2605.16953v1 Announce Type: new Abstract: While AI-generated hallucinations pose considerable risks, the underlying cognitive mechanisms by which humans can successfully recognize or be misled by these hallucinations remain unclear. To address this problem, this paper...

<https://arxiv.org/abs/2605.16953>

1 CONF

Harnessing AI for Inverse Partial Differential Equation Problems: Past, Present, and Prospects

ArXiv CS.AI 50%

arXiv:2605.16966v1 Announce Type: new Abstract: Solving inverse partial differential equation (PDE) problems is a fundamental topic in scientific research due to its broad significance across a wide range of real-world applications. Inverse PDE problems arise across medical...

<https://arxiv.org/abs/2605.16966>

2 CONF

Brain Vascular Age Prediction Using Cerebral Blood Flow Velocity and Machine Learning Algorithms

ArXiv CS.AI 50%

arXiv:2605.16969v1 Announce Type: new Abstract: Defining vascular age in terms of physiological function has become one focal point of the extensive studies to categorize and track chronological age. Transcranial Doppler (TCD) is a method by which cerebral blood flow velocity is...

<https://arxiv.org/abs/2605.16969>

3 **CONF****Reliability and Effectiveness of Autonomous AI Agents in Supply Chain Management****ArXiv CS.AI** **50%**

arXiv:2605.17036v1 Announce Type: new Abstract: This paper studies autonomous generative AI agents in multi-echelon supply chains using the MIT Beer Game. We identify four inference-time levers that shape performance: model selection, policies and guardrails, centralized data...

<https://arxiv.org/abs/2605.17036>

4 **CONF****AnchorDiff: Topology-Aware Masked Diffusion with Confidence-based Rewriting for Radiology Report Generation****ArXiv CS.AI** **50%**

arXiv:2605.17071v1 Announce Type: new Abstract: Radiology report generation (RRG) aims to automatically produce clinically accurate textual reports from medical images. Existing methods predominantly rely on autoregressive (AR) language models, whose causal dependency structure...

<https://arxiv.org/abs/2605.17071>

5 **CONF****ConflictRAG: Detecting and Resolving Knowledge Conflicts in Retrieval Augmented Generation****ArXiv CS.AI** **50%**

arXiv:2605.17301v1 Announce Type: cross Abstract: Retrieval-Augmented Generation (RAG) systems implicitly assume mutual consistency among retrieved documents -- an assumption that frequently fails in practice. We present ConflictRAG, a conflict-aware RAG framework that detects,...

<https://arxiv.org/abs/2605.17301>

6 **CONF****Scientific Logicality Enriched Methodology for LLM Reasoning: A Practice in Physics****ArXiv CS.AI** **50%**

arXiv:2605.17104v1 Announce Type: new Abstract: With the continuous advancement of reasoning abilities in Large Language Models (LLMs), their application to scientific reasoning tasks has gained significant research attention. Current research primarily emphasizes boosting LLMs'...

<https://arxiv.org/abs/2605.17104>

7 **CONF****Capturing LLM Capabilities via Evidence-Calibrated Query Clustering****ArXiv CS.AI** **50%**

arXiv:2605.17110v1 Announce Type: new Abstract: Query clustering organizes queries into groups that reflect shared latent capability demands, enabling capability-aware LLM evaluation. Existing clustering methods, which primarily rely on semantic taxonomies or embeddings, often...

<https://arxiv.org/abs/2605.17110>

8 **CONF****F2IND-IT! -- Multimodal Fuzzy Fake Indian News Detection using Images and Text****ArXiv CS.AI** **50%**

arXiv:2605.17115v1 Announce Type: new Abstract: Biased manipulation of facts across regional and national media outlets complicates misinformation detection in diverse landscapes like India. This paper introduces a novel multimodal framework combining visual and textual...

<https://arxiv.org/abs/2605.17115>

9 CONF

Dynamics of collective creativity in AI art competitions

ArXiv CS.AI 50%

arXiv:2605.17141v1 Announce Type: new Abstract: Creativity is a fundamental aspect of how culture evolves, yet the mechanisms by which groups produce novelty are notoriously difficult to infer from the historical record. Iterated learning experiments have shown that cultural...

<https://arxiv.org/abs/2605.17141>

0 CONF

MADP: A Multi-Agent Pipeline for Sustainable Document Processing with Human-in-the-Loop

ArXiv CS.AI 50%

arXiv:2605.17159v1 Announce Type: new Abstract: Document processing automation remains a critical challenge in enterprise environments, where traditional manual approaches are labor-intensive and error-prone. We present MADP, a multi-agent architecture that addresses the...

<https://arxiv.org/abs/2605.17159>

1 CONF

CheeseBench: Evaluating Large Language Models on Rodent Behavioral Neuroscience Paradigms

ArXiv CS.AI 50%

arXiv:2604.10825v2 Announce Type: replace Abstract: We introduce CheeseBench, a benchmark that evaluates large language models (LLMs) on nine classical behavioral neuroscience paradigms (Morris water maze, Barnes maze, T-maze, radial arm maze, star maze, operant chamber, shuttle...

<https://arxiv.org/abs/2604.10825>

2 CONF

CoLLM-NAS: Collaborative Large Language Models for Efficient Knowledge-Guided Neural Architecture Search

ArXiv CS.AI 50%

arXiv:2509.26037v2 Announce Type: replace Abstract: The integration of Large Language Models (LLMs) with Neural Architecture Search (NAS) has introduced new possibilities for automating the design of neural architectures. However, most existing methods face critical limitations,...

<https://arxiv.org/abs/2509.26037>

3 CONF

CatalyticMLLM: A Graph-Text Multimodal Large Language Model for Catalytic Materials

ArXiv CS.AI 50%

arXiv:2605.17254v1 Announce Type: new Abstract: Property prediction and inverse structural design of catalytic materials are typically modeled as two independent tasks: the former predicts target properties from given structures, whereas the latter generates candidate structures...

<https://arxiv.org/abs/2605.17254>

4 CONF

Is VLA Reasoning Faithful? Probing Safety of Chain-of-Causation

ArXiv CS.AI 50%

arXiv:2605.17268v1 Announce Type: new Abstract: We present the first systematic study of faithfulness in Vision-Language-Action (VLA) driving models, analyzing 300 Alpamayo-R1-10B inferences across 100 diverse PhysicalAI-AV scenarios. Our main finding is that output...

<https://arxiv.org/abs/2605.17268>

- 5
CONF
A2RBench: An Automatic Paradigm for Formally Verifiable Abstract Reasoning Benchmark Generation
ArXiv CS.AI 50%
arXiv:2605.17278v1 Announce Type: new Abstract: Abstract reasoning ability reflects the intelligence and generalization capacity of LLMs to extract and apply abstract rules. However, accurately measuring this ability remains challenging: existing benchmarks either rely on...
<https://arxiv.org/abs/2605.17278>
- 6
CONF
MetaCogAgent: A Metacognitive Multi-Agent LLM Framework with Self-Aware Task Delegation
ArXiv CS.AI 50%
arXiv:2605.17292v1 Announce Type: new Abstract: Multi-agent large language model (LLM) systems have shown promise for solving complex tasks through agent collaboration. However, existing frameworks assign tasks based on predefined roles without considering whether an agent can...
<https://arxiv.org/abs/2605.17292>
- 7
CONF
Agentic AI Translate: An Agentic Translator Prototype for Translation as Communication Design
ArXiv CS.AI 50%
arXiv:2605.17041v1 Announce Type: cross Abstract: We present Agentic AI Translate, an agentic translator prototype that operationalises the thesis of Yamada (forthcoming) -- that the metalanguage of Translation Studies has become an instruction code for generative AI. The system...
<https://arxiv.org/abs/2605.17041>
- 8
CONF
QQJ: Quantifying Qualitative Judgment for Scalable and Human-Aligned Evaluation of Generative AI
ArXiv CS.AI 50%
arXiv:2605.17382v1 Announce Type: new Abstract: The rapid progress of generative artificial intelligence has exposed fundamental limitations in existing evaluation methodologies, particularly for open-ended, creative, and human-facing tasks. Traditional automatic metrics rely on...
<https://arxiv.org/abs/2605.17382>
- 9
CONF
Computational Challenges in Token Economics: Bridging Economic Theory and AI System Design
ArXiv CS.AI 50%
arXiv:2605.17410v1 Announce Type: new Abstract: Token economics has emerged as a useful lens for understanding resource allocation, value creation, and pricing in large language model systems. While recent work has increasingly treated tokens as economic primitives, there...
<https://arxiv.org/abs/2605.17410>
- 0
CONF
RAG-based EEG-to-Text Translation Using Deep Learning and LLMs
ArXiv CS.AI 50%
arXiv:2605.17503v1 Announce Type: new Abstract: The decoding of linguistic information from electroencephalography (EEG) signals remains an extremely challenging problem in brain-computer interface (BCI) research. In particular, sentence-level decoding from EEG is difficult due...
<https://arxiv.org/abs/2605.17503>

1 **CONF****Memory-Guided Tree Search with Cross-Branch Knowledge Transfer for LLM Solver Synthesis****ArXiv CS.AI 50%**

arXiv:2605.17539v1 Announce Type: new Abstract: Combinatorial optimization (CO) underlies decision-making from logistics to chip design, where infeasible solutions are operationally unusable and small quality gains translate into substantial economic value. Recent work uses...

<https://arxiv.org/abs/2605.17539>

2 **CONF****Evaluating Deep Research Agents on Expert Consulting Work: A Benchmark with Verifiers, Rubrics, and Cognitive Traps****ArXiv CS.AI 50%**

arXiv:2605.17554v1 Announce Type: new Abstract: Frontier deep research agents (DRAs) plan a research task, synthesize across documents, and return a structured deliverable on demand. They are being deployed in enterprise workflows faster than they are being evaluated. Existing...

<https://arxiv.org/abs/2605.17554>

3 **CONF****Reverse-Engineering Model Editing on Language Models****ArXiv CS.AI 50%**

arXiv:2602.10134v2 Announce Type: replace-cross Abstract: Large language models (LLMs) are pretrained on corpora containing trillions of tokens and, therefore, inevitably memorize sensitive information. Locate-then-edit methods, as a mainstream paradigm of model editing, offer a...

<https://arxiv.org/abs/2602.10134>

4 **CONF****SocialMemBench: Are AI Memory Systems Ready for Social Group Settings?****ArXiv CS.AI 50%**

arXiv:2605.17789v1 Announce Type: cross Abstract: Memory systems for AI assistants were built for single-user dialogue and fail characteristically when applied to multi-party social group settings. This gap matters for the social assistants being built today: group-acting agents...

<https://arxiv.org/abs/2605.17789>

5 **CONF****AutoRubric-T2I: Robust Rule-Based Reward Model for Text-to-Image Alignment****ArXiv CS.AI 50%**

arXiv:2605.17602v1 Announce Type: new Abstract: Aligning Text-to-Image (T2I) generation models with human preferences increasingly relies on image reward models that score or rank generated images according to prompt alignment and perceptual quality. Existing reward models are...

<https://arxiv.org/abs/2605.17602>

6 **CONF****KIT-TIP-NLP at MultiPride: Continual Learning with Multilingual Foundation Model****ArXiv CS.AI 50%**

arXiv:2605.13415v2 Announce Type: replace-cross Abstract: This paper presents a multi-stage framework for detecting reclaimed slurs in multilingual social media discourse. It addresses the challenge of identifying reclamatory versus non-reclamatory usage of LGBTQ+-related slurs...

<https://arxiv.org/abs/2605.13415>

7 **CONF****A Machine Learning Framework for EEG-Based Prediction of Treatment Efficacy in Chronic Neck Pain****ArXiv CS.AI** **50%**

arXiv:2605.16326v1 Announce Type: cross Abstract: Chronic neck pain is a leading cause of disability worldwide, and current treatment selection remains largely trial and error. We present a machine learning framework that uses electroencephalography to predict treatment efficacy...

<https://arxiv.org/abs/2605.16326>

8 **CONF****Agents for Experiments, Experiments for Agents: A Design Grammar for AI-Enabled Experimental Science****ArXiv CS.AI** **50%**

arXiv:2605.17746v1 Announce Type: new Abstract: AI systems are becoming active participants in organizational and knowledge work. They increasingly interact with humans, coordinate workflows, and operate in multi-agent arrangements. Understanding their effects therefore requires...

<https://arxiv.org/abs/2605.17746>

9 **CONF****Accelerating AI-Powered Research: The PuppyChatter Framework for Usable and Flexible Tooling****ArXiv CS.AI** **50%**

arXiv:2605.17809v1 Announce Type: new Abstract: This research addresses the challenges inherent in developing Artificial Intelligence (AI) applications, particularly those leveraging Large Language Models (LLMs). While AI vendors provide Application Programming Interfaces (APIs)...

<https://arxiv.org/abs/2605.17809>

0 **CONF****Going Headless? On the Boundaries of Vertical AI Firms****ArXiv CS.AI** **50%**

arXiv:2605.17812v1 Announce Type: new Abstract: Vertical AI firms in accounting, law, healthcare, procurement, and similar domains historically bundled workflow, domain logic, and accountability into a single application. General-purpose AI agents are now unbundling that...

<https://arxiv.org/abs/2605.17812>

1 **CONF****AgentArk: Distilling Multi-Agent Intelligence into a Single LLM Agent****ArXiv CS.AI** **50%**

arXiv:2602.03955v2 Announce Type: replace Abstract: While large language model (LLM) multi-agent systems achieve superior reasoning performance through iterative debate, practical deployment is limited by their high computational cost and error propagation. This paper proposes...

<https://arxiv.org/abs/2602.03955>

2 **CONF****UniversalRAG: Retrieval-Augmented Generation over Corpora of Diverse Modalities and Granularities****ArXiv CS.AI** **50%**

arXiv:2504.20734v4 Announce Type: replace-cross Abstract: Retrieval-Augmented Generation (RAG) has shown substantial promise in improving factual accuracy by grounding model responses with external knowledge relevant to queries. However, most existing approaches are limited to a...

<https://arxiv.org/abs/2504.20734>

- 3

CONF

Agentic Chunking and Bayesian De-chunking of AI Generated Fuzzy Cognitive Maps: A Model of the Thucydides Trap

ArXiv CS.AI

50%

arXiv:2605.17903v1 Announce Type: new Abstract: We automatically generate feedback causal fuzzy cognitive maps (FCMs) from text by teaching large-language-model agents to break the text into overlapping chunks of text. Convex mixing of these chunk FCMs gives a representative...

<https://arxiv.org/abs/2605.17903>

4

CONF

DataClawBench: An Agent Benchmark for Exploratory Real-World Financial Data Analysis

ArXiv CS.AI

50%

arXiv:2605.02503v2 Announce Type: replace Abstract: Autonomous data analysis agents are increasingly expected to conduct exploratory analysis over underexplored data environments. This burden is especially salient in complex financial analytics, where relevant evidence is rarely...

<https://arxiv.org/abs/2605.02503>

5

CONF

Rethinking Code Review in the Age of AI: A Vision for Agentic Code Review

ArXiv CS.AI

50%

arXiv:2605.17548v1 Announce Type: cross Abstract: Code review has evolved for decades, from informal peer checking to today's pull request (PR) workflows, yet it remains a largely manual, uneven, and cognitively demanding process. The rise of Artificial Intelligence (AI) coding...

<https://arxiv.org/abs/2605.17548>

6

CONF

Dynamic Generation of Multi-LLM Agents Communication Topologies with Graph Diffusion Models

ArXiv CS.AI

50%

arXiv:2510.07799v2 Announce Type: replace-cross Abstract: The efficiency of multi-agent systems driven by large language models (LLMs) largely hinges on their communication topology. However, designing an optimal topology is a non-trivial challenge, as it requires balancing...

<https://arxiv.org/abs/2510.07799>

7

CONF

CAM-VFD: Cross-Attention Multimodal Video Forgery Detection

ArXiv CS.AI

50%

arXiv:2605.17133v1 Announce Type: cross Abstract: The rapid advancement of Deepfake technologies and video manipulation tools poses a critical challenge to multimedia forensics, judicial evidence integrity, and information authenticity. Current detectors rely on single-modality...

<https://arxiv.org/abs/2605.17133>

8

CONF

Stabilizing Temporal Inference Dynamics for Online Surgical Phase Recognition

ArXiv CS.AI

50%

arXiv:2605.16387v1 Announce Type: cross Abstract: Online Surgical Phase Recognition (SPR) models can reach high frame-wise accuracy, yet their predictions often lack temporal stability, fragmenting workflow understanding and reducing the reliability of downstream assistance. We...

<https://arxiv.org/abs/2605.16387>

9 CONF

Generative AI and Two-Tiered Online Mental Health Communities

ArXiv CS.AI 50%

arXiv:2605.16279v1 Announce Type: cross Abstract: Online mental health communities (OMHCs) are tiered platforms that connect patients with licensed counselors through public Q&A forums and paid private consultations. Their two-tier structure creates a strategic dilemma for genAI...

<https://arxiv.org/abs/2605.16279>

0 CONF

Visualizing the Invisible: Generative Visual Grounding Empowers Universal EEG Understanding in MLLMs

ArXiv CS.AI 50%

arXiv:2605.18172v1 Announce Type: new Abstract: Leveraging the universal representations of pre-trained LLMs and MLLMs offers a promising path toward brain foundation models. However, visually-evoked EEG datasets remain scarce, leading existing methods to align neural signals...

<https://arxiv.org/abs/2605.18172>

1 CONF

RoboMME: Benchmarking and Understanding Memory for Robotic Generalist Policies

ArXiv CS.AI 50%

arXiv:2603.04639v2 Announce Type: replace-cross Abstract: Memory is critical for long-horizon and history-dependent robotic manipulation. Such tasks often involve counting repeated actions or manipulating objects that become temporarily occluded. Recent vision-language-action...

<https://arxiv.org/abs/2603.04639>

2 CONF

Metric-Guided Feature Fusion of Visual Foundation Models for Segmentation Tasks

ArXiv CS.AI 50%

arXiv:2605.16864v1 Announce Type: cross Abstract: Although large-scale visual foundation models (VFM) achieve remarkable performance in semantic understanding, they still underperform in instance-aware dense prediction tasks. They exhibit different biases in representation: for...

<https://arxiv.org/abs/2605.16864>

3 CONF

MirrorBench: A Benchmark to Evaluate Conversational User-Proxy Agents for Human-Likeness

ArXiv CS.AI 50%

arXiv:2601.08118v3 Announce Type: replace Abstract: Large language models (LLMs) are increasingly used as human simulators, both for evaluating conversational systems and for generating fine-tuning data. However, naive "act-as-a-user" prompting often yields verbose, unrealistic...

<https://arxiv.org/abs/2601.08118>

4 CONF

Query-Conditioned Knowledge Alignment for Reliable Cross-System Medical Reasoning

ArXiv CS.AI 50%

arXiv:2605.18570v1 Announce Type: new Abstract: Cross-domain knowledge alignment is essential for integrating heterogeneous medical systems, yet existing approaches typically treat entity alignment as a static matching problem, ignoring query context and cross-system asymmetry....

<https://arxiv.org/abs/2605.18570>

- 5 **CONF**
- When Outcome Looks Right But Discipline Fails: Trace-Based Evaluation Under Hidden Competitor State**
- ArXiv CS.AI 50%
- arXiv:2605.18580v1 Announce Type: new Abstract: Outcome-only evaluation can certify economically unsafe agents: a policy can hit a business KPI while violating deployable behavioral discipline. In hotel pricing with hidden competitor state, a learner can achieve plausible...
- <https://arxiv.org/abs/2605.18580>
-
- 6 **CONF**
- Position: A Three-Layer Probabilistic Assume-Guarantee Architecture Is Structurally Required for Safe LLM Agent Deployment**
- ArXiv CS.AI 50%
- arXiv:2605.18672v1 Announce Type: new Abstract: This position paper argues that enforcing LLM agent safety within a single abstraction layer is not merely suboptimal but categorically insufficient for deployed LLM agents -- a structural consequence of how agent execution works,...
- <https://arxiv.org/abs/2605.18672>
-
- 7 **CONF**
- Democratizing Large-Scale Re-Optimization with LLM-Guided Model Patches**
- ArXiv CS.AI 50%
- arXiv:2605.18692v1 Announce Type: new Abstract: Optimization models developed by operations research (OR) experts are often deployed as decision-support systems in industrial settings. However, real-world environments are dynamic, with evolving business rules, previously...
- <https://arxiv.org/abs/2605.18692>
-
- 8 **CONF**
- Friends and Grandmothers in Silico: Localizing Entity Cells in Language Models**
- ArXiv CS.AI 50%
- arXiv:2604.01404v2 Announce Type: replace-cross Abstract: How do language models retrieve entity-specific facts from their parameters? We investigate this question by searching for sparse, entity-selective MLP neurons - which we call entity cells, by analogy to the "grandmother..."
- <https://arxiv.org/abs/2604.01404>
-
- 9 **CONF**
- SIPO: Stabilized and Improved Preference Optimization for Aligning Diffusion Models**
- ArXiv CS.AI 50%
- arXiv:2505.21893v3 Announce Type: replace-cross Abstract: Preference learning has garnered extensive attention as an effective technique for aligning diffusion models with human preferences in visual generation. However, existing alignment approaches such as Diffusion-DPO suffer...
- <https://arxiv.org/abs/2505.21893>
-
- 0 **CONF**
- Helping Customers in Distress: An LLM-powered Agent that Converses, Probes, and Routes**
- ArXiv CS.AI 50%
- arXiv:2605.16268v1 Announce Type: cross Abstract: Banks receive millions of reports of fraud, scams, and disputed transactions every year, making it challenging to accurately direct customers to the appropriate specialist teams for assistance. The existing manual process driven...
- <https://arxiv.org/abs/2605.16268>
-

1 CONF

SENSE: Satellite-based ENergy Synthesis for Sustainable Environment

ArXiv CS.AI 50%

arXiv:2605.18101v1 Announce Type: cross Abstract: Urban Building Energy Modeling plays a critical role in achieving the United Nations' Sustainable Development Goals 7 and 11. Although existing studies based on satellite imagery and deep learning have achieved remarkable...

<https://arxiv.org/abs/2605.18101>

2 CONF

Retrieval and competition: how a protein foundation model starts a protein

ArXiv CS.AI 50%

arXiv:2605.16331v1 Announce Type: cross Abstract: Protein language models are increasingly used to guide experimental and clinical decisions, yet it is often unclear whether a confident prediction reflects recognition of biological evidence or retrieval of a statistical default....

<https://arxiv.org/abs/2605.16331>

3 CONF

ChartDesign: Towards LLM Designer of Data Visualization

ArXiv CS.AI 50%

arXiv:2605.16274v1 Announce Type: cross Abstract: Charts are the dominant medium for visualizing data, discovering patterns and trends, and communicating data driven insights, yet designing them still requires expensive human effort and expertise, such as selecting appropriate...

<https://arxiv.org/abs/2605.16274>

4 CONF

AI Slop or AI-enhancement? Student perceptions of AI-generated media for an English for Academic Purposes course

ArXiv CS.AI 50%

arXiv:2605.16275v1 Announce Type: cross Abstract: Artificial intelligence (AI) retrieval-augmented generation (RAG) tools now enable educators to transform course materials into diverse multimedia at scale. However, it remains unclear whether such AI-generated content functions...

<https://arxiv.org/abs/2605.16275>

5 CONF

Generative AI in K-12 Classrooms: A Midyear Implementation Report

ArXiv CS.AI 50%

arXiv:2605.16277v1 Announce Type: cross Abstract: This mid-year report summarizes teacher use of Colleague AI across 12 Washington State school districts from September 1 to December 31, 2025. Produced jointly by Colleague AI and AmplifyLearn.AI at the University of Washington,...

<https://arxiv.org/abs/2605.16277>

6 CONF

Keeping an Eye on AI: A Framework for Effective Human Oversight of AI Systems

ArXiv CS.AI 50%

arXiv:2605.16278v1 Announce Type: cross Abstract: The use of Artificial Intelligence (AI) in high-risk, decision-making scenarios presents technical, safety, and normative challenges; problems that may only be ameliorated by human oversight. However, notions of human oversight...

<https://arxiv.org/abs/2605.16278>

7 CONF

From Reactive to Proactive: A Multi-Regulatory Empirical Analysis of 480 AI Incidents and a Data-Driven Governance Compliance Framework

ArXiv CS.AI 50%

arXiv:2605.16281v1 Announce Type: cross Abstract: Artificial intelligence systems are increasingly deployed in high-stakes domains, yet it remains unclear whether existing governance frameworks ensure accountability after deployment. This study makes two contributions. First, it...

<https://arxiv.org/abs/2605.16281>

8 CONF

AI of the People, by the People, for the People: A Social Choice Approach to Collective Control of Artificial Intelligence

ArXiv CS.AI 50%

arXiv:2605.16291v1 Announce Type: cross Abstract: With the growing adoption of AI systems, reasoning about how society can exert control over AI becomes an increasingly urgent problem. Existing work on democratic control largely focuses on macro-level governance. In contrast, we...

<https://arxiv.org/abs/2605.16291>

9 CONF

Evidence of a Cognitive Shift in AI Education: How Students Are Rethinking Human Intelligence?

ArXiv CS.AI 50%

arXiv:2605.16292v1 Announce Type: cross Abstract: Perceptions of intelligence shape how learners evaluate and rely on artificial intelligence (AI) systems. Despite rapid advances in AI capabilities, the impact of sustained exposure to these tools on students' valuation of human...

<https://arxiv.org/abs/2605.16292>

0 CONF

From Demographics to Survey Anchors: Evaluating LLM Agents for Modeling Retirement Attitudes

ArXiv CS.AI 50%

arXiv:2605.16303v1 Announce Type: cross Abstract: Large language models (LLM) agents may offer tools to predict human responses to surveys. A common technique for defining these agents uses only demographics, for example country, age, gender, employment status, income, education...

<https://arxiv.org/abs/2605.16303>

1 CONF

ACE: Self-Evolving LLM Coding Framework via Adversarial Unit Test Generation and Preference Optimization

ArXiv CS.AI 50%

arXiv:2605.16299v1 Announce Type: cross Abstract: Large Language Models (LLMs) excel at code generation but remain heavily reliant on large-scale annotated solutions and verification-based supervision, which constrains scalability and hinders sustained self-improvement. Recent...

<https://arxiv.org/abs/2605.16299>

2 CONF

Consent Chain Degradation in Embodied Multi-Agent Systems: Bridging the Gap Between AI Agent Governance and Robot Ethics

ArXiv CS.AI 50%

arXiv:2605.16300v1 Announce Type: cross Abstract: Robotic systems are moving from isolated platforms to interconnected multi-agent ecosystems that operate in human environments. This shift raises a governance problem that existing frameworks do not address: how does consent...

<https://arxiv.org/abs/2605.16300>

3 **CONF****UVTran: Accurate Hole-Filling Parameterization with Transformers****ArXiv CS.AI** **50%**

arXiv:2605.16306v1 Announce Type: cross Abstract: In industrial design, N-sided hole filling is typically formulated as the construction of a single trimmed B-spline surface by minimizing a fairness energy subject to geometric boundary constraints. This formulation requires an...

<https://arxiv.org/abs/2605.16306>

4 **CONF****A Structural Threshold in Decision Capacity Governs Collapse in Self-Play Reinforcement Learning****ArXiv CS.AI** **50%**

arXiv:2605.16315v1 Announce Type: cross Abstract: We show that a threshold in decision capacity determines whether self-play reinforcement learning agents collapse under asymmetric rule perturbations. Across poker variants, matrix games, a dice game, and multiple learning...

<https://arxiv.org/abs/2605.16315>

5 **CONF****Detecting Verbatim LLM Copy-Paste in Homework****ArXiv CS.AI** **50%**

arXiv:2605.16336v1 Announce Type: cross Abstract: Large language models (LLMs) have made fluent essay writing, code drafting, and quiz answering instantly available to students at every level, from secondary school through graduate study. Many educators do not object to LLM use...

<https://arxiv.org/abs/2605.16336>

6 **CONF****Federated Nested Learning: Collaborative Training of Self-Referential Memories for Test-Time Adaptation****ArXiv CS.AI** **50%**

arXiv:2605.16350v1 Announce Type: cross Abstract: We rethink Federated Learning (FL) from a nested learning perspective, framing the core challenge as how to collaboratively learn optimization rules, not just static models, to tackle Non-IID client data. To address this, we...

<https://arxiv.org/abs/2605.16350>

7 **CONF****Augmenting Human Evaluation with LLM Judges: How Many Human Reviews Do You Need?****ArXiv CS.AI** **50%**

arXiv:2605.16354v1 Announce Type: cross Abstract: Large language models (LLMs) are increasingly used as automated evaluators of AI systems, including in high-stakes applications. In this role, LLMs are used to generate judgments about the quality, appropriateness, or even safety...

<https://arxiv.org/abs/2605.16354>

8 **CONF****LEAF: A Living Benchmark for Event-Augmented Forecasting****ArXiv CS.AI** **50%**

arXiv:2605.16358v1 Announce Type: cross Abstract: Large Language Models (LLMs) are increasingly applied to forecasting. To evaluate this capability while mitigating pre-training data contamination, several living benchmarks have been proposed. However, existing benchmarks either...

<https://arxiv.org/abs/2605.16358>

9 CONF

ProxyKV: Cross-Model Proxy Pruning for Efficient Long-Context LLM Inference

ArXiv CS.AI 50%

arXiv:2605.16360v1 Announce Type: cross Abstract: Efficient long-context inference in Large Language Models (LLMs) is severely constrained by the Key-Value (KV) cache memory wall, yet existing pruning methods force a choice between low-latency heuristics that sacrifice precision...

<https://arxiv.org/abs/2605.16360>

00 CONF

GPU-Accelerated Deep Learning for Heatwave Prediction and Urban Heat Risk Assessment

ArXiv CS.AI 50%

arXiv:2605.16435v1 Announce Type: cross Abstract: Heatwaves are an important problem in cities, and climate change makes this problem more difficult. In this paper, we present a GPU-based deep learning framework for next-day prediction of urban thermal conditions and for heat...

<https://arxiv.org/abs/2605.16435>

01 CONF

WASIL: In-the-Wild Arabic Spoken Interactions with LLMs

ArXiv CS.AI 50%

arXiv:2605.16364v1 Announce Type: cross Abstract: Large Language Models (LLMs) voice assistants are commonly built as cascaded Automatic Speech recognition (ASR) to LLM systems, where recognition errors can distort user intent. Dislikes may also arise from ambiguous,...

<https://arxiv.org/abs/2605.16364>

02 CONF

RaBiT: Residual-Aware Binarization Training for Accurate and Efficient LLMs

ArXiv CS.AI 50%

arXiv:2602.05367v2 Announce Type: replace Abstract: Efficient deployment of large language models (LLMs) requires extreme quantization, forcing a critical trade-off between low-bit efficiency and performance. Residual binarization enables hardware-friendly, matmul-free inference...

<https://arxiv.org/abs/2602.05367>

03 CONF

GeoSymb127K: Scalable Symbolically-verifiable Synthesis for Multimodal Geometric Reasoning

ArXiv CS.AI 50%

arXiv:2605.16371v1 Announce Type: cross Abstract: Large Multimodal Models (LMMs) often struggle with geometric reasoning due to visual hallucinations and a lack of mathematically precise Chain-of-Thought (CoT) data. To address this, we propose the GeoSymb Engine, an automated and...

<https://arxiv.org/abs/2605.16371>

04 CONF

CheckSupport: A Local LLM-Powered Tool for Automated Manuscript Submission Checklist Selection and Completion

ArXiv CS.AI 50%

arXiv:2605.16377v1 Announce Type: cross Abstract: Transparent and standardized reporting is essential for reproducible scientific research, yet adherence to reporting guidelines remains inconsistent because of the manual effort required to select and complete checklists. We...

<https://arxiv.org/abs/2605.16377>

05 CONF

Probing Persona-Dependent Preferences in Language Models

ArXiv CS.AI 50%

arXiv:2605.13339v2 Announce Type: replace-cross Abstract: Large language models (LLMs) can be said to have preferences: they reliably pick certain tasks and outputs over others, and preferences shaped by post-training and system prompts appear to shape much of their behaviour....

<https://arxiv.org/abs/2605.13339>

06 CONF

A neurosymbolic Approach with Epistemic Deep Learning for Hierarchical Image Classification

ArXiv CS.AI 50%

arXiv:2605.16383v1 Announce Type: cross Abstract: Deep neural networks achieve high accuracy on image classification tasks. Yet, they often produce overconfident predictions as which fail to express epistemic uncertainty, and frequently violate logical or structural constraints...

<https://arxiv.org/abs/2605.16383>

07 CONF

One-Block Transformer (1BT) for EEG-Based Cognitive Workload Assessment

ArXiv CS.AI 50%

arXiv:2605.00856v2 Announce Type: replace-cross Abstract: Accurate and continuous estimation of cognitive workload is fundamental to creating adaptive human-machine systems. However, designing architectures that balance representational capacity with computational efficiency has...

<https://arxiv.org/abs/2605.00856>

08 CONF

Trajectory-Aware Adaptive Inference in Object Detection Models

ArXiv CS.AI 50%

arXiv:2605.16397v1 Announce Type: cross Abstract: The increasing integration of sensors in autonomous maritime navigation has led to large-scale multimodal datasets, raising challenges in achieving efficient real-time perception. In such systems, object detection and trajectory...

<https://arxiv.org/abs/2605.16397>

09 CONF

Neural Visual Decoding via Cognitive guided Adaptive Blurring and Information Constrained Alignment

ArXiv CS.AI 50%

arXiv:2605.16418v1 Announce Type: cross Abstract: EEG-based visual decoding aims to establish a mapping between neural signals and visual semantics. However, it remains constrained by the dual challenges of severe information granularity mismatch and the low signal-to-noise...

<https://arxiv.org/abs/2605.16418>

10 CONF

The Impact of AI Search on the Online Content Ecosystem: Evidence from Google and Reddit

ArXiv CS.AI 50%

arXiv:2605.16428v1 Announce Type: cross Abstract: Search engines traditionally complement online content platforms by directing users seeking information to external websites. The emergence of generative AI search tools that summarize answers directly on the results page may...

<https://arxiv.org/abs/2605.16428>

11 CONF

QuantFPFlow: Quantum Amplitude Estimation for Fokker-Planck Policy Optimisation in Continuous Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2605.16429v1 Announce Type: cross Abstract: We introduce QuantFPFlow , a reinforcement learning framework that integrates quantum amplitude estimation into the Fokker-Planck (FP) formulation of stochastic policy optimisation. Classical continuous-space RL agents...

<https://arxiv.org/abs/2605.16429>

12 CONF

Edge-AI-Driven Learning-to-Rank for Decentralized Task Allocation in Circular Smart Manufacturing

ArXiv CS.AI 50%

arXiv:2605.16433v1 Announce Type: cross Abstract: Task allocation in smart manufacturing systems needs to operate under decentralized decision-making, dynamic workloads, and shared resource constraints. In circular manufacturing settings, these challenges are further intensified...

<https://arxiv.org/abs/2605.16433>

13 CONF

Transitivity Meets Cyclicity: Explicit Preference Decomposition for Dynamic Large Language Model Alignment

ArXiv CS.AI 50%

arXiv:2605.17342v1 Announce Type: cross Abstract: Standard RLHF relies on transitive scalar rewards, failing to capture the cyclic nature of human preferences. While some approaches like the General Preference Model (GPM) address this, we identify a theoretical limitation: their...

<https://arxiv.org/abs/2605.17342>

14 CONF

Diffusion Attention Expert Model for Predicting and Semi-automatic Localizing STAS in Lung Cancer Histopathological Images

ArXiv CS.AI 50%

arXiv:2605.16444v1 Announce Type: cross Abstract: Accurate intraoperative and postoperative diagnosis of spread through air spaces (STAS) is essential for guiding surgical decisions and postoperative management in lung cancer. However, histopathological assessment is...

<https://arxiv.org/abs/2605.16444>

15 CONF

Physics-Guided Geometric Diffusion for Macro Placement Generation

ArXiv CS.AI 50%

arXiv:2605.16451v1 Announce Type: cross Abstract: Macro placement is a pivotal stage in VLSI physical design, fundamentally determining the overall chip performance. Recent data-driven placement methods have demonstrated significant potential, yet they often struggle to handle...

<https://arxiv.org/abs/2605.16451>

16 CONF

LEAP: Learnable End-to-End Adaptive Pruning of Large Language Models

ArXiv CS.AI 50%

arXiv:2605.17289v1 Announce Type: cross Abstract: Unstructured sparsity is now natively accelerated by recent GPU kernels and dataflow hardware, shifting the bottleneck from inference execution to the pruning algorithm. State-of-the-art methods for unstructured LLM pruning are...

<https://arxiv.org/abs/2605.17289>

17 CONF

Conservative AI for Safety-Sensitive Medical Image Restoration: Residual-Bounded CT-CTA Enhancement for Intracranial Aneurysm-Relevant Signal Recovery

ArXiv CS.AI 50%

arXiv:2605.16458v1 Announce Type: cross Abstract: Image restoration models are increasingly applied to degraded medical scans, but in safety-sensitive settings they must improve image quality without uncontrolled modification of clinically important regions. This is especially...

<https://arxiv.org/abs/2605.16458>

18 CONF

Mechanistically Interpretable Neural Encoding Reveals Fine-Grained Functional Selectivity in Human Visual Cortex

ArXiv CS.AI 50%

arXiv:2605.16468v1 Announce Type: cross Abstract: A central goal in understanding human vision is to uncover the visual features that drive neuronal activity. A growing body of work has used artificial neural networks as encoding models to predict cortical responses to natural...

<https://arxiv.org/abs/2605.16468>

19 CONF

Learning from Historical Activations in Graph Neural Networks

ArXiv CS.AI 50%

arXiv:2601.01123v2 Announce Type: replace-cross Abstract: Graph Neural Networks (GNNs) have demonstrated remarkable success in various domains such as social networks, molecular chemistry, and more. A crucial component of GNNs is the pooling procedure, in which the node features...

<https://arxiv.org/abs/2601.01123>

20 CONF

No Plan, Yet Human: A Reactive Robotics Model Predicts Human Planning Failures on a Clinical Task

ArXiv CS.AI 50%

arXiv:2605.16514v1 Announce Type: cross Abstract: Understanding why some sequential planning problems are harder than others requires models that go beyond average performance. They should capture the specific pattern of which problems are hard, and ideally fail in the same way...

<https://arxiv.org/abs/2605.16514>

21 CONF

Alignment Drift in Long-Term Human-LLM Interaction: A Mechanism-Oriented Framework

ArXiv CS.AI 50%

arXiv:2605.16516v1 Announce Type: cross Abstract: Long-term interaction with LLM-based systems may produce alignment drift: a gradual process in which system outputs become less constrained by the user's current message and more shaped by prior interaction history, while still...

<https://arxiv.org/abs/2605.16516>

22 CONF

Toward Template-Free Explainability for Monte Carlo Tree Search

ArXiv CS.AI 50%

arXiv:2605.16524v1 Announce Type: cross Abstract: Probabilistic search algorithms, such as Monte Carlo Tree Search (MCTS), have proven very effective in solving sequential decision-making tasks under uncertainty. However, interpreting asymmetric search trees that incorporate...

<https://arxiv.org/abs/2605.16524>

23 CONF

Inventorship in AI-Assisted Inventions: Designing an Experiment to Shape Case Law

ArXiv CS.AI 50%

arXiv:2605.16528v1 Announce Type: cross Abstract: The latest improvements in artificial intelligence (AI) raise new challenges for intellectual property laws, particularly concerning the inventorship issue in AI-assisted inventions - that is, those in which AI is used in the...

<https://arxiv.org/abs/2605.16528>

24 CONF

How Few-Shot Examples Add Up: A Causal Decomposition of Function Vectors in In-Context Learning

ArXiv CS.AI 50%

arXiv:2605.16591v1 Announce Type: cross Abstract: In-context learning (ICL) excels at new tasks from minimal examples, yet we still lack a mechanistic explanation of how few-shot prompts shape a model's function vector (FV)--a causal activation direction that drives task...

<https://arxiv.org/abs/2605.16591>

25 CONF

Where Pretraining writes and Alignment reads: the asymmetry of Transformer weight space

ArXiv CS.AI 50%

arXiv:2605.16600v1 Announce Type: cross Abstract: Cross-entropy pretraining and preference alignment update the same transformer weights, but leave geometrically distinct traces. We characterise this asymmetry with a relative-subspace-fraction probe that tracks how weight deltas...

<https://arxiv.org/abs/2605.16600>

26 CONF

Why Modeling Human Haptic Material Perception with AI Is Difficult

ArXiv CS.AI 50%

arXiv:2605.16602v1 Announce Type: cross Abstract: Touch plays a central role in how humans perceive and recognize materials through physical contact. Despite decades of research, the mechanisms by which tactile signals are transformed into meaningful perceptual representations...

<https://arxiv.org/abs/2605.16602>

27 CONF

To Trust or Not to Trust: Authors' Response to AI-based Reviews

ArXiv CS.AI 50%

arXiv:2605.16623v1 Announce Type: cross Abstract: Large language models are increasingly discussed and used as tools that may assist with scholarly peer review, but empirical evidence regarding how authors use and perceive AI-based feedback remains limited. This paper reports...

<https://arxiv.org/abs/2605.16623>

28 CONF

LivePI: More Realistic Benchmarking of Agents Against Indirect Prompt Injectio

ArXiv CS.AI 50%

arXiv:2605.17986v1 Announce Type: cross Abstract: AI agents such as OpenClaw are increasingly deployed in local workflows with access to external tools. This creates indirect prompt-injection (IPI) risk: an agent may execute harmful instructions embedded in untrusted inputs such...

<https://arxiv.org/abs/2605.17986>

29 CONF

Multi-Object Tracking Consistently Improves Wildlife Inference

ArXiv CS.AI 50%

arXiv:2605.16672v1 Announce Type: cross Abstract: Camera traps have become a common tool for wildlife monitoring efforts in ecological research and biodiversity conservation. Wildlife classification models have benefited from the increase in wildlife visual data. These models...

<https://arxiv.org/abs/2605.16672>

30 CONF

CHI-Bench: Can AI Agents Automate End-to-End, Long-Horizon, Policy-Rich Healthcare Workflows?

ArXiv CS.AI 50%

arXiv:2605.16679v1 Announce Type: cross Abstract: End-to-end automation of realistic healthcare operations stresses three capabilities underrepresented in current benchmarks: policy density, decisions must be grounded in a large library of medical, insurance, and operational...

<https://arxiv.org/abs/2605.16679>

31 CONF

EmoMind: Decoding Affective Captions from Human Brain fMRI

ArXiv CS.AI 50%

arXiv:2605.16739v1 Announce Type: cross Abstract: Decoding visual experience from brain activity has advanced substantially, but current brain-to-text systems largely recover semantic content while discarding affect. Additionally, language models can generate emotional text...

<https://arxiv.org/abs/2605.16739>

32 CONF

BlendedNet++: A dataset and benchmark for field-resolved aerodynamics and inverse design of blended wing body aircraft

ArXiv CS.AI 50%

arXiv:2512.03280v2 Announce Type: replace-cross Abstract: The conceptual design of Blended Wing Body (BWB) aircraft is often constrained by the high computational cost of resolving complex aerodynamics over a high-dimensional design space. While deep learning offers a pathway to...

<https://arxiv.org/abs/2512.03280>

33 CONF

VolTA-3D: Self-Supervised Learning for Brain MRI using 3D Volumetric Token Alignment

ArXiv CS.AI 50%

arXiv:2605.16775v1 Announce Type: cross Abstract: Self-supervised learning (SSL) has advanced medical image analysis by enabling learning from large unlabelled data. However, in brain magnetic resonance imaging (MRI), most 3D models remain specialized for either segmentation of...

<https://arxiv.org/abs/2605.16775>

34 CONF

Cross-Domain Molecular Relational Learning: Leveraging Chemical Structure-Activity Analysis

ArXiv CS.AI 50%

arXiv:2605.16799v1 Announce Type: cross Abstract: Recent advances in molecular representation integrates molecular topological and visual modalities, opening new avenues for precise Molecular Relational Learning (MRL). Existing MRL methods focus on intra-domain modeling, and...

<https://arxiv.org/abs/2605.16799>

35 CONF

Cross-modal Affinity-aligned Multimodal Learning Analytics for Predicting Student Collaboration Satisfaction in Game-Based Learning

ArXiv CS.AI 50%

arXiv:2605.16806v1 Announce Type: cross Abstract: Collaborative game-based learning environments offer rich opportunities for small-group knowledge construction, yet automatically predicting student collaboration satisfaction remains challenging. A critical barrier is modality...

<https://arxiv.org/abs/2605.16806>

36 CONF

AgentKernelArena: Generalization-Aware Benchmarking of GPU Kernel Optimization Agents

ArXiv CS.AI 50%

arXiv:2605.16819v1 Announce Type: cross Abstract: GPU kernel optimization is increasingly critical for efficient deep learning systems, but writing high-performance kernels still requires substantial low-level expertise. Recent AI coding agents can iteratively read code, invoke...

<https://arxiv.org/abs/2605.16819>

37 CONF

When Marginals Match but Structure Fails: Covariance Fidelity in Generative Models

ArXiv CS.AI 50%

arXiv:2603.17041v2 Announce Type: replace-cross Abstract: Generative models are increasingly deployed as substitutes for real data in downstream scientific workflows, yet standard evaluation criteria remain focused on marginal distribution matching. We argue that this represents...

<https://arxiv.org/abs/2603.17041>

38 CONF

Decoupling KL and Trajectories: A Unified Perspective for SFT, DAgger, Offline RL, and OPD in LLM Distillation

ArXiv CS.AI 50%

arXiv:2605.16826v1 Announce Type: cross Abstract: Knowledge distillation is central to LLM post-training, yet its design space remains poorly understood, especially alongside reinforcement learning (RL). We show that the prevailing paradigms, off-policy distillation and...

<https://arxiv.org/abs/2605.16826>

39 CONF

Plan First, Diffuse Later: Extrinsic Graph Guidance for Long-Horizon Diffusion Planning

ArXiv CS.AI 50%

arXiv:2605.16863v1 Announce Type: cross Abstract: Compositional diffusion models offer a promising route to long-horizon planning by denoising multiple overlapping sub-trajectories while ensuring that together they constitute a global solution. However, enforcing local behavior...

<https://arxiv.org/abs/2605.16863>

40 CONF

Some[Body] Must Receive That Pain for Agent Accountability

ArXiv CS.AI 50%

arXiv:2605.16872v1 Announce Type: cross Abstract: AI agents increasingly act consequentially in the real world. This creates a problem we call \emph{consequence reception}: harm occurs, the producing system is identified, yet no continuing agent receives consequences in a way...

<https://arxiv.org/abs/2605.16872>

41 CONF

The Alpha Illusion: Reported Alpha from LLM Trading Agents Should Not Be Treated as Deployment Evidence

ArXiv CS.AI 50%

arXiv:2605.16895v1 Announce Type: cross Abstract: End-to-end LLM trading agents have moved quickly from research curiosity to a small ecosystem of named systems, including FinCon, FinMem, TradingAgents, FinAgent, QuantAgent, and FLAG-Trader. Several of these report headline...

<https://arxiv.org/abs/2605.16895>

42 CONF

Full Attention Strikes Back: Transferring Full Attention into Sparse within Hundred Training Steps

ArXiv CS.AI 50%

arXiv:2605.16928v1 Announce Type: cross Abstract: Long-context inference in large language models is bottlenecked by the quadratic cost of full attention. Existing efficient alternatives often rely either on native sparse training or on heuristic token eviction, creating an...

<https://arxiv.org/abs/2605.16928>

43 CONF

Effort as Ceiling, Not Dial: Reasoning Budget Does Not Modulate Cognitive Cost Alignment Between Humans and Large Reasoning Models

ArXiv CS.AI 50%

arXiv:2605.16938v1 Announce Type: cross Abstract: Large Reasoning Models (LRMs) generate chain-of-thought traces whose length tracks human reaction times across cognitive tasks, but recent debate questions whether this alignment reflects genuine computational structure or...

<https://arxiv.org/abs/2605.16938>

44 CONF

WhiteTesseract: Reframing the Interpretation of Cultural Heritage through XR and Conversational AI

ArXiv CS.AI 50%

arXiv:2605.16972v1 Announce Type: cross Abstract: Cultural heritage exhibitions often struggle to sustain attention and support reflective engagement. Physical exhibitions rely on fixed interpretive aids that lack adaptability to individual backgrounds or curiosity, and their...

<https://arxiv.org/abs/2605.16972>

45 CONF

Skills on the Fly: Test-Time Adaptive Skill Synthesis for LLM Agents

ArXiv CS.AI 50%

arXiv:2605.16986v1 Announce Type: cross Abstract: LLM agents benefit from reusable skills, yet test-time tasks often require guidance more specific than a static skill library can provide. We propose \emph{SkillTTA}, a Test-Time Adaptive Skill Synthesis method that retrieves a...

<https://arxiv.org/abs/2605.16986>

46 CONF

Response-free item difficulty modelling for multiple-choice items with fine-tuned transformers: Component-wise representation and multi-task learning

ArXiv CS.AI 50%

arXiv:2605.16991v1 Announce Type: cross Abstract: Response-free item difficulty modelling promises to reduce reliance on response-based calibration but is intrinsically difficult on reading-comprehension multiple-choice items, where difficulty depends on inferential demands...

<https://arxiv.org/abs/2605.16991>

47 CONF

Adversarial Fragility and Language Vulnerability in Clinical AI: A Systematic Audit of Diagnostic Collapse Under Imperceptible Perturbations and Cross-Lingual Drift in Low-Resource Healthcare Settings

ArXiv CS.AI 50%

arXiv:2605.16993v1 Announce Type: cross Abstract: Current clinical artificial intelligence (AI) systems are evaluated almost exclusively on clean, standardised, English-language inputs, conditions that do not reflect the realities of healthcare delivery in low-resource settings....

<https://arxiv.org/abs/2605.16993>

48 CONF

Learning-Zone Energy: Online Data Selection for Efficient RL Post-Training

ArXiv CS.AI 50%

arXiv:2605.17003v1 Announce Type: cross Abstract: Reinforcement Learning (RL) post-training has emerged as the dominant paradigm for eliciting mathematical reasoning in Large Language Models (LLMs), yet prevailing techniques such as GRPO and DAPO distribute rollout and gradient...

<https://arxiv.org/abs/2605.17003>

49 CONF

When Dynamics Shift, Robust Task Inference Wins: Offline Imitation Learning with Behavior Foundation Models Revisited

ArXiv CS.AI 50%

arXiv:2605.17017v1 Announce Type: cross Abstract: Behavior Foundation Models (BFMs) enable scalable imitation learning (IL) by pretraining task-agnostic representations that can be rapidly adapted to new tasks. However, existing BFMs assume fixed environment dynamics, limiting...

<https://arxiv.org/abs/2605.17017>

50 CONF

1GC-7RC: One Graphic Card -- Seven Research Challenges! How Good Are AI Agents at Doing Your Job?

ArXiv CS.AI 50%

arXiv:2605.17046v1 Announce Type: cross Abstract: Autonomous AI coding agents are becoming a core tool for ML practitioners in industry and research alike. Despite this growing adoption, no standardized benchmark exists to evaluate their ability to design, implement, and train...

<https://arxiv.org/abs/2605.17046>

51 CONF

Can LLMs Think Like Consumers? Benchmarking Crowd-Level Reaction Reconstruction with ConsumerSimBench

ArXiv CS.AI 50%

arXiv:2605.17079v1 Announce Type: cross Abstract: LLMs are increasingly used as "digital consumers" to simulate public opinion, pre-test marketing decisions, and anticipate audience response. However, existing evaluations rarely ask whether a model can reconstruct the concrete...

<https://arxiv.org/abs/2605.17079>

52 CONF

Visual Timelines of Police Encounters in Body-Worn Camera Footage: Operational Context and Activity Cataloging for Training and Analysis in OpenBWC

ArXiv CS.AI 50%

arXiv:2605.17095v1 Announce Type: cross Abstract: Law enforcement agencies are accumulating vast amounts of body-worn camera (BWC) footage. However, this remains operationally opaque. That is, analysts and trainers still have to invest considerable time watching full-length...

<https://arxiv.org/abs/2605.17095>

53 CONF

UCSF-PDGM-VQA: Visual Question Answering dataset for brain tumor MRI interpretation

ArXiv CS.AI 50%

arXiv:2605.17140v1 Announce Type: cross Abstract: Brain tumor diagnosis is largely dependent on Magnetic Resonance Imaging (MRI) evaluation, which requires radiologists to synthesize thousands of images across multiple 3D sequences and longitudinal studies. This process requires...

<https://arxiv.org/abs/2605.17140>

54 CONF

When Bits Break Recourse: Counterfactual-Faithful Quantization

ArXiv CS.AI 50%

arXiv:2605.17160v1 Announce Type: cross Abstract: Quantization can preserve predictive accuracy under low-bit deployment while silently breaking algorithmic recourse: an actionable change that flips a decision before quantization may fail after quantization, or become...

<https://arxiv.org/abs/2605.17160>

55 CONF

OpenJarvis: Personal AI, On Personal Devices

ArXiv CS.AI 50%

arXiv:2605.17172v1 Announce Type: cross Abstract: Personal AI stacks, like OpenClaw and Hermes Agent, are becoming central to daily work, yet they route nearly every query (often over sensitive local data) to cloud-hosted frontier models. Replacing frontier models with local...

<https://arxiv.org/abs/2605.17172>

56 CONF

PluRule: A Benchmark for Moderating Pluralistic Communities on Social Media

ArXiv CS.AI 50%

arXiv:2605.17187v1 Announce Type: cross Abstract: Social media are shifting towards pluralism -- community-governed platforms where groups define their own norms. What violates rules in one community may be perfectly acceptable in another. Can AI models help moderate such...

<https://arxiv.org/abs/2605.17187>

57 CONF

Integration of AI in Cybersecurity: Current Trends with a Focused Look at Intrusion Detection Applications

ArXiv CS.AI 50%

arXiv:2605.17219v1 Announce Type: cross Abstract: Artificial Intelligence (AI) is widely adopted today for its ability to detect patterns, automate tasks, and reduce time and cost across various applications. Its integration into Cybersecurity has garnered significant attention,...

<https://arxiv.org/abs/2605.17219>

58 CONF

Systematic Evaluation of Vision Transformers for Automated Cervical Cancer Classification: Optimization, Statistical Validation, and Clinical Interpretability

ArXiv CS.AI 50%

arXiv:2605.17236v1 Announce Type: cross Abstract: Manual Pap smear analysis for cervical cancer screening is limited by inter-observer variability, time constraints, and restricted expert availability. Although convolutional neural networks (CNNs) have automated cervical cell...

<https://arxiv.org/abs/2605.17236>

59 **CONF****Latency-Aware Deep Learning Benchmark for Real-Time Cyber-Physical Attack and Fault Classification in Inverter-Dominated Power Grids**

ArXiv CS.AI 50%

arXiv:2605.17256v1 Announce Type: cross Abstract: This work introduces a latency-aware benchmarking framework for evaluating deep learning models in power system anomaly detection using high-fidelity, time-domain signals generated from an industry-grade electromagnetic transient...

<https://arxiv.org/abs/2605.17256>

60 **CONF****State-of-the-Art Claims Require State-of-the-Art Evidence**

ArXiv CS.AI 50%

arXiv:2605.17273v1 Announce Type: cross Abstract: State-of-the-Art (SOTA) claims pervade Artificial Intelligence (AI) and Machine Learning (ML) research. These claims rest on benchmark evaluations, where models are ranked by aggregate scores across tasks. Public benchmarks or...

<https://arxiv.org/abs/2605.17273>

61 **CONF****Rover: Context-aware Conflict Resolution with LLM**

ArXiv CS.AI 50%

arXiv:2605.17279v1 Announce Type: cross Abstract: Code merging is a significant challenge, particularly in large-scale projects. Existing solutions, including program analysis and machine learning, show promise but face critical limitations. Program analysis lacks the ability to...

<https://arxiv.org/abs/2605.17279>

62 **CONF****ContractBench: Can LLM Agents Preserve Observation Contracts?**

ArXiv CS.AI 50%

arXiv:2605.17281v1 Announce Type: cross Abstract: Tool-augmented LLM agents call APIs whose intermediate outputs, such as presigned URLs, session tokens, and OAuth state parameters, are observation contracts: artifacts whose later use is constrained by the external system that...

<https://arxiv.org/abs/2605.17281>

63 **CONF****When Efficiency Backfires: Cascading LLMs Trigger Cascade Failure under Adversarial Attack**

ArXiv CS.AI 50%

arXiv:2605.17288v1 Announce Type: cross Abstract: Large Language Model (LLM) cascade systems are designed to balance efficiency and performance by processing queries with lightweight models while selectively escalating complex cases to more powerful ones. Such systems seek to...

<https://arxiv.org/abs/2605.17288>

64 **CONF****Forgetting is Competition: Rethinking Unlearning as Representation Interference in Diffusion Models**

ArXiv CS.AI 50%

arXiv:2603.00975v2 Announce Type: replace-cross Abstract: Deployed text-to-image diffusion models increasingly require post-hoc concept unlearning for copyright claims, artist opt-outs, safety updates, and protected-content mitigation without full retraining. A central challenge...

<https://arxiv.org/abs/2603.00975>

65 CONF

ASPI: Seeking Ambiguity Clarification Amplifies Prompt Injection Vulnerability in LLM Agents

ArXiv CS.AI 50%

arXiv:2605.17324v1 Announce Type: cross Abstract: Clarification-seeking behavior is widely regarded as a desirable property of LLM agents, enabling them to resolve ambiguity before acting on underspecified tasks. However, the security implications of this interaction pattern...

<https://arxiv.org/abs/2605.17324>

66 CONF

Single-Sample Black-Box Membership Inference Attack against Vision-Language Models via Cross-modal Semantic Alignment

ArXiv CS.AI 50%

arXiv:2605.17341v1 Announce Type: cross Abstract: Vision-Language Models (VLMs) have achieved remarkable success, yet their reliance on massive datasets and unintended memorization of training data raise significant data security risk. Membership Inference Attacks (MIAs) aim to...

<https://arxiv.org/abs/2605.17341>

67 CONF

FML-bench: A Controlled Study of AI Research Agent Strategies from the Perspective of Search Dynamics

ArXiv CS.AI 50%

arXiv:2605.17373v1 Announce Type: cross Abstract: AI research agents accelerate ML research by automating hypothesis generation, experimentation, and empirical refinement. Existing agent strategies range from greedy hill-climbing to tree search and evolutionary optimization, yet...

<https://arxiv.org/abs/2605.17373>

68 CONF

Benchmarking Mythos-Linked Bug Rediscovery

ArXiv CS.AI 50%

arXiv:2605.17416v1 Announce Type: cross Abstract: Anthropic's April 2026 Mythos materials combine benchmark claims with concrete bug-finding stories across OpenBSD, FreeBSD, Linux, FFmpeg, and browsers. This paper reports a controlled target-file rediscovery experiment on six...

<https://arxiv.org/abs/2605.17416>

69 CONF

Learning Displacement-Robust Representations for Landslide Early Warning under Rainfall Forecast Uncertainty

ArXiv CS.AI 50%

arXiv:2605.17419v1 Announce Type: cross Abstract: Rainfall-induced landslides pose a growing risk worldwide as climate change intensifies extreme rainfall events. To provide sufficient evacuation time, landslide early warning systems (LEWS) for real-time disaster monitoring must...

<https://arxiv.org/abs/2605.17419>

70 CONF

Progressive Generalization Augmentation with Deeply Coupled RND-PPO and Domain-Prioritized Noise Injection for Robust Crop Management Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2605.17428v1 Announce Type: cross Abstract: Our preliminary experiments on gym-DSSAT maize irrigation tasks revealed that +/-2 degrees C temperature noise causes an 11.9% reduction in economic returns for PPO policies trained under clean conditions - a systematic...

<https://arxiv.org/abs/2605.17428>

71 CONF

MATE: Solving Contextual Markov Decision Processes with Memory of Accumulated Transition Embeddings

ArXiv CS.AI 50%

arXiv:2605.17431v1 Announce Type: cross Abstract: We propose MATE, a simple yet effective memory architecture for solving Contextual Markov Decision Processes (CMDPs), a family of MDPs parameterized by an unobserved context. In CMDPs, an optimal agent can adapt online by...

<https://arxiv.org/abs/2605.17431>

72 CONF

Beyond Catalogue Counts: the Dataset Visibility Asymmetry in Low-Resource Multilingual NLP

ArXiv CS.AI 50%

arXiv:2605.17442v1 Announce Type: cross Abstract: Multilingual NLP often relies on dataset counts from centralized catalogues to characterize which languages are resource-rich or resource-poor. However, these catalogues record only one layer of dataset visibility: what has been...

<https://arxiv.org/abs/2605.17442>

73 CONF

MemRepair: Hierarchical Memory for Agentic Repository-Level Vulnerability Repair

ArXiv CS.AI 50%

arXiv:2605.17444v1 Announce Type: cross Abstract: Modern software ecosystems face a rapidly growing number of disclosed vulnerabilities, increasing the need for automated repair techniques that can operate reliably at repository scale. Although Large Language Model (LLM)-based...

<https://arxiv.org/abs/2605.17444>

74 CONF

ContraFix: Agentic Vulnerability Repair via Differential Runtime Evidence and Skill Reuse

ArXiv CS.AI 50%

arXiv:2605.17450v1 Announce Type: cross Abstract: Large language model (LLM) agents are increasingly used for automated vulnerability repair (AVR), where repository-level reasoning enables them to inspect context and produce source-code patches. However, recent empirical results...

<https://arxiv.org/abs/2605.17450>

75 CONF

Artificial Intelligence can Recognize Whether a Job Applicant is Selling and/or Lying According to Facial Expressions and Head Movements Much More Correctly Than Human Interviewers

ArXiv CS.AI 50%

arXiv:2605.17461v1 Announce Type: cross Abstract: Whether an interviewee's honest and deceptive responses can be detected by facial expression signals in videos has been debated and requires further research. We developed deep learning models enabled by computer vision to...

<https://arxiv.org/abs/2605.17461>

76 CONF

An Interpretable Closed-Loop Intelligent Tutoring System for Multimodal Affective Feedback in Asynchronous Presentation Training

ArXiv CS.AI 50%

arXiv:2605.17468v1 Announce Type: cross Abstract: This paper presents an interpretable closed-loop Intelligent Tutoring System (ITS) that supports feedback-guided practice for developing on-camera oral presentation skills at scale. The system operationalizes a seven-dimensional...

<https://arxiv.org/abs/2605.17468>

77 CONF

Beyond Linear Superposition: Discovering Climate Features in AI Weather Models with KAN-SAE

ArXiv CS.AI 50%

arXiv:2605.17493v1 Announce Type: cross Abstract: Deep learning weather prediction models achieve remarkable predictive skill yet remain largely opaque: we know little about how they represent physical climate phenomena internally. Mechanistic interpretability through Sparse...

<https://arxiv.org/abs/2605.17493>

78 CONF

A Distributional View for Visual Mechanistic Interpretability: KL-Minimal Soft-Constraint Principle

ArXiv CS.AI 50%

arXiv:2605.17504v1 Announce Type: cross Abstract: Most current paradigms in visual mechanistic interpretability (MI) remain confined to interpreting internal units of the vision model via heuristic methods (e.g., top-\$K\$ activation retrieval or optimization with regularization)....

<https://arxiv.org/abs/2605.17504>

79 CONF

Beyond Accuracy: Robustness, Interpretability and Expressiveness of EEG Foundation Models

ArXiv CS.AI 50%

arXiv:2605.17562v1 Announce Type: cross Abstract: EEG foundation models (EEG-FMs) have been evaluated predominantly on clean, in-distribution accuracy, leaving their robustness, interpretability and representational quality largely unexamined. This study addresses these gaps by...

<https://arxiv.org/abs/2605.17562>

80 CONF

LLMForge: Multi-Backend Hardware-Aware Neural Architecture Search with Infinite-Head Attention for Edge Language Models

ArXiv CS.AI 50%

arXiv:2605.17653v1 Announce Type: cross Abstract: Sub-billion-parameter Transformer language models are increasingly deployed on edge devices, where the privacy, latency, and operating-cost advantages of on-device inference are constrained by tight memory-bandwidth, energy, and...

<https://arxiv.org/abs/2605.17653>

81 CONF

Beyond Correctness: Harmonizing Process and Outcome Rewards through RL Training

ArXiv CS.AI 50%

arXiv:2509.03403v2 Announce Type: replace-cross Abstract: Reinforcement Learning with Verifiable Rewards (RLVR) improves final-answer accuracy on reasoning tasks, but it does not reliably improve reasoning quality. Because outcome rewards only assess final answers, they also...

<https://arxiv.org/abs/2509.03403>

82 CONF

Validate Your Authority: Benchmarking LLMs on Multi-Label Precedent Treatment Classification

ArXiv CS.AI 50%

arXiv:2605.17691v1 Announce Type: cross Abstract: Automating the classification of negative treatment in legal precedent is a critical yet nuanced NLP task where misclassification carries significant risk. To address the shortcomings of standard accuracy, this paper introduces a...

<https://arxiv.org/abs/2605.17691>

83 **CONF****Geometry-Aware Attention Guidance for Diffusion Models via Modern Hopfield Dynamics**

ArXiv CS.AI 50%

arXiv:2603.02531v2 Announce Type: replace-cross Abstract: Classifier-Free Guidance (CFG) improves sample quality in diffusion models, but its dual-pass inference and reliance on null-condition training limit its use in few-step regimes. Attention-space guidance has emerged as a...

<https://arxiv.org/abs/2603.02531>

84 **CONF****Bridging the Version Gap: Multi-version Training Improves ICD Code Prediction, Especially for Rare Codes**

ArXiv CS.AI 50%

arXiv:2605.17755v1 Announce Type: cross Abstract: Clinical coding maps clinical documentation to standardized medical codes, an essential yet time-consuming administrative task that could benefit from automation. Current models on ICD coding are typically optimized for codes...

<https://arxiv.org/abs/2605.17755>

85 **CONF****Systematic Evaluation of the Quality of Synthetic Clinical Notes Rephrased by LLMs at Million-Note Scale**

ArXiv CS.AI 50%

arXiv:2605.17775v1 Announce Type: cross Abstract: Large language models (LLMs) can generate or synthesize clinical text for a wide range of applications, from improving clinical documentation to augmenting clinical text analytics. Yet evaluations typically focus on a narrow...

<https://arxiv.org/abs/2605.17775>

86 **CONF****One Model, Two Roles: Emergent Specialization in a Shared Recurrent Transformer**

ArXiv CS.AI 50%

arXiv:2605.17811v1 Announce Type: cross Abstract: Can a shared-weight recurrent Transformer develop distinct internal roles without being partitioned into separate modules? We study this in Asymmetric Input Recurrence (AIR), a minimal two-state reasoning architecture in which...

<https://arxiv.org/abs/2605.17811>

87 **CONF****Why We Look Where We Look: Emergent Human-like Fixations of a Foveated Visual Language Model Maximizing Scene Understanding**

ArXiv CS.AI 50%

arXiv:2605.17823v1 Announce Type: cross Abstract: When humans view scenes without a specific task (free-viewing), they initially direct their eye movements toward the scene center and then fixate on people, text, objects being gazed at or grasped, and semantically meaningful...

<https://arxiv.org/abs/2605.17823>

88 **CONF****DARC: Disagreement-Aware Alignment via Risk-Constrained Decoding**

ArXiv CS.AI 50%

arXiv:2603.08145v2 Announce Type: replace-cross Abstract: Preference-based alignment methods (e.g., RLHF, DPO) typically optimize a single scalar objective, implicitly averaging over heterogeneous human preferences. In practice, systematic annotator and user-group disagreement...

<https://arxiv.org/abs/2603.08145>

89 CONF

Babel: Jailbreaking Safety Attention via Obfuscation Distribution Optimized Sampling

ArXiv CS.AI 50%

arXiv:2605.17971v1 Announce Type: cross Abstract: Despite rigorous safety alignment, Large Language Models (LLMs) remain vulnerable to jailbreak attacks. Existing black-box methods often rely on heuristic templates or exhaustive trials, lacking mechanistic interpretability and...

<https://arxiv.org/abs/2605.17971>

90 CONF

SAFE-SVD: Sensitivity-Aware Fidelity-Enforcing SVD for Physics Foundation Models

ArXiv CS.AI 50%

arXiv:2605.17985v1 Announce Type: cross Abstract: We propose a new method for compressing physics foundation models (PFMs) which is a new trend in AI for Science. While model compression is essential for reducing memory use and accelerating inference in large foundation models,...

<https://arxiv.org/abs/2605.17985>

91 CONF

SignRoundV2: Toward Closing the Performance Gap in Extremely Low-Bit Post-Training Quantization for LLMs

ArXiv CS.AI 50%

arXiv:2512.04746v2 Announce Type: replace-cross Abstract: Extremely low-bit quantization is critical for efficiently deploying Large Language Models (LLMs), yet it often leads to severe performance degradation at 2 bits and even at 4 bits (e.g., MXFP4). We present SignRoundV2, a...

<https://arxiv.org/abs/2512.04746>

92 CONF

Spiker-LL: An Energy-Efficient FPGA Accelerator Enabling Adaptive Local Learning in Spiking Neural Networks

ArXiv CS.AI 50%

arXiv:2605.18003v1 Announce Type: cross Abstract: Deploying adaptive intelligence at the edge remains challenging due to the high computational and energy cost of training neural models. Spiking Neural Networks (SNNs) offer a promising alternative, but enabling on-device...

<https://arxiv.org/abs/2605.18003>

93 CONF

Quantum Sidecar Architectures for Hybrid AI Training and Inference: Stateful Protected Registers, Stateless Reset-and-Reprepare Circuits and Quantum Weight-State Outlook

ArXiv CS.AI 50%

arXiv:2605.18031v1 Announce Type: cross Abstract: We propose a quantum sidecar architecture family for future hybrid AI training and inference. The central idea is not to store an entire Transformer in a small quantum memory, nor to claim one-shot collapse into a fully trained...

<https://arxiv.org/abs/2605.18031>

94 CONF

Exploring Trust Calibration in XAI - The Impact of Exposing Model Limitations to Lay Users

ArXiv CS.AI 50%

arXiv:2605.18036v1 Announce Type: cross Abstract: Trust calibration -- aligning user trust judgment with model capability -- is crucial for safe deployment of explainable AI (XAI), yet is often evaluated via global trust ratings detached from objective performance evidence. We...

<https://arxiv.org/abs/2605.18036>

95 **CONF****Confidence-Gated Robot Autonomy: When Does Uncertainty Actually Help?**

ArXiv CS.AI 50%

arXiv:2605.18045v1 Announce Type: cross Abstract: Robotic systems often use predictive uncertainty to decide whether to act autonomously or defer to a fallback policy. In threshold-gated autonomy, uncertainty matters mainly through its ability to rank likely errors. Standard...

<https://arxiv.org/abs/2605.18045>
96 **CONF****FLAG: Foundation model representation with Latent diffusion Alignment via Graph for spatial gene expression prediction**

ArXiv CS.AI 50%

arXiv:2605.18055v1 Announce Type: cross Abstract: Predicting spatial gene expression from routine H&E enables large-scale molecular profiling, yet current models treat this as isolated pointwise tasks, thereby overlooking essential biological structures like gene coordination...

<https://arxiv.org/abs/2605.18055>
97 **CONF****Symmetry-Compatible Principle for Optimizer Design: Embeddings, LM Heads, SwiGLU MLPs, and MoE Routers**

ArXiv CS.AI 50%

arXiv:2605.18106v1 Announce Type: cross Abstract: A striking geometric disparity has long persisted in the practice of deep learning. While modern neural network architectures naturally exhibit rich symmetry and equivariance properties, popular optimizers such as Adam and its...

<https://arxiv.org/abs/2605.18106>
98 **CONF****An Empirical Study of Privacy Leakage Chains via Prompt Injection in Black-Box Chatbot Environments**

ArXiv CS.AI 50%

arXiv:2605.18133v1 Announce Type: cross Abstract: LLM-based chatbot agents increasingly process user requests by combining natural-language reasoning with external tools such as web browsing. These capabilities improve usability, but they also create attack surfaces when...

<https://arxiv.org/abs/2605.18133>
99 **CONF****PIPER: Content-Based Table Search via profiling and LLM-Generated Pseudoqueries**

ArXiv CS.AI 50%

arXiv:2605.18199v1 Announce Type: cross Abstract: The rapid growth of tabular datasets in data lakes, data spaces, and open data portals makes effective dataset search essential for reuse and analysis. Existing search systems rely mainly on metadata, which is often incomplete or...

<https://arxiv.org/abs/2605.18199>
00 **CONF****Leveraging Graph Structure in Seq2Seq Models for Knowledge Graph Link Prediction**

ArXiv CS.AI 50%

arXiv:2605.18211v1 Announce Type: cross Abstract: We introduce Graph-Augmented Sequence-to-Sequence (GA-S2S), a novel framework that integrates a T5-small encoder-decoder with a Relational Graph Attention Network (RGAT) to improve link prediction in knowledge graphs. While...

<https://arxiv.org/abs/2605.18211>

01 CONF

Are Sparse Autoencoder Benchmarks Reliable?

ArXiv CS.AI 50%

arXiv:2605.18229v1 Announce Type: cross Abstract: Sparse autoencoders (SAEs) are a core interpretability tool for large language models, and progress on SAE architectures depends on benchmarks that reliably distinguish better SAEs from worse ones. We audit the SAE quality...

<https://arxiv.org/abs/2605.18229>

02 CONF

SomaliWeb v1: A Quality-Filtered Somali Web Corpus with a Matched Tokenizer and a Public Language-Identification Benchmark

ArXiv CS.AI 50%

arXiv:2605.18232v1 Announce Type: cross Abstract: Somali is a Cushitic language of the Horn of Africa with ~25 million speakers, yet no documented dedicated Somali pretraining corpus with a companion tokenizer and language-identification benchmark has been publicly released....

<https://arxiv.org/abs/2605.18232>

03 CONF

Toward Robust Multilingual Adaptation of LLMs for Low-Resource Languages

ArXiv CS.AI 50%

arXiv:2510.14466v3 Announce Type: replace-cross Abstract: Large language models (LLMs) continue to struggle with low-resource languages, primarily due to limited training data, translation noise, and unstable cross-lingual alignment. To address these challenges, we propose LiRA...

<https://arxiv.org/abs/2510.14466>

04 CONF

Privacy Preserving Reinforcement Learning with One-Sided Feedback

ArXiv CS.AI 50%

arXiv:2605.18246v1 Announce Type: cross Abstract: We study reinforcement learning (RL) in multi-dimensional continuous state and action spaces with one-sided feedback, where the agent receives partial observations of the state and obtains reward information for only a subset of...

<https://arxiv.org/abs/2605.18246>

05 CONF

From Volume to Value: Preference-Aligned Memory Construction for On-Device RAG

ArXiv CS.AI 50%

arXiv:2605.18271v1 Announce Type: cross Abstract: With the rapid emergence of personal AI agents based on Large Language Models (LLMs), implementing them on-device has become essential for privacy and responsiveness. To handle the inherently personal and context-dependent nature...

<https://arxiv.org/abs/2605.18271>

06 CONF

PH-Dreamer: A Physics-Driven World Model via Port-Hamiltonian Generative Dynamics

ArXiv CS.AI 50%

arXiv:2605.18303v1 Announce Type: cross Abstract: World models built on recurrent state space architectures enable efficient latent imagination, yet remain physically unstructured, producing dynamics that violate conservation and dissipative principles. We introduce a unified...

<https://arxiv.org/abs/2605.18303>

07 CONF

Probing SMEFT Operators through $t\bar{t}\bar{t}$ Production with Hyper-Graph Neural Networks at the LHC

ArXiv CS.AI 50%

arXiv:2605.18382v1 Announce Type: cross Abstract: We present a phenomenological study of $t\bar{t}\bar{t}$ production in proton-proton collisions at $\sqrt{s} = 13\text{--}14\text{ TeV}$, using a Hyper-Graph Neural Network (H-GNN) to discriminate multilepton signal events from the dominant SM...

<https://arxiv.org/abs/2605.18382>

08 CONF

Diagnosing Korean-Language LLM Political Bias via Census-Grounded Agent Simulation

ArXiv CS.AI 50%

arXiv:2605.18395v1 Announce Type: cross Abstract: Large language models (LLMs) exhibit systematic political biases in voter simulations, but their underlying mechanisms and cross-lingual generalizations remain poorly understood. We introduce Dynamo-K, a census-grounded...

<https://arxiv.org/abs/2605.18395>

09 CONF

Qumus: Realization of An Embodied AI Quantum Material Experimentalist

ArXiv CS.AI 50%

arXiv:2605.18407v1 Announce Type: cross Abstract: While modern Large Language Models (LLMs) and agentic artificial intelligence (AI) have demonstrated transformative capabilities in digital domains, the realization of embodied AI capable of real-world scientific discovery...

<https://arxiv.org/abs/2605.18407>

10 CONF

An Assessment of Human vs. Model Uncertainty in Soft-Label Learning and Calibration

ArXiv CS.AI 50%

arXiv:2605.18648v1 Announce Type: cross Abstract: Central to human-aligned AI is understanding the benefits of human-elicited labels over synthetic alternatives. While human soft-labels improve calibration by capturing uncertainty, prior studies conflate these benefits with the...

<https://arxiv.org/abs/2605.18648>

11 CONF

EvoMemBench: Benchmarking Agent Memory from a Self-Evolving Perspective

ArXiv CS.AI 50%

arXiv:2605.18421v1 Announce Type: cross Abstract: Recent benchmarks for Large Language Model (LLM) agents mainly evaluate reasoning, planning, and execution. However, memory is also essential for agents, as it enables them to store, update, and retrieve information over time....

<https://arxiv.org/abs/2605.18421>

12 CONF

Prompt2Fingerprint: Plug-and-Play LLM Fingerprinting via Text-to-Weight Generation

ArXiv CS.AI 50%

arXiv:2605.18474v1 Announce Type: cross Abstract: The widespread deployment and redistribution of large language models (LLMs) have made model provenance tracking a critical challenge. While existing LLM fingerprinting methods, particularly active approaches that embed identity...

<https://arxiv.org/abs/2605.18474>

13 CONF

AI4BayesCode: From Natural Language Descriptions to Validated Modular Stateful Bayesian Samplers

ArXiv CS.AI 50%

arXiv:2605.18476v1 Announce Type: cross Abstract: Coding and computation remain major bottlenecks in Markov chain Monte Carlo (MCMC) workflows, especially as modern sampling algorithms have become increasingly complex and existing probabilistic programming systems remain limited...

<https://arxiv.org/abs/2605.18476>

14 CONF

Continuous Diffusion Scales Competitively with Discrete Diffusion for Language

ArXiv CS.AI 50%

arXiv:2605.18530v1 Announce Type: cross Abstract: While diffusion has drawn considerable recent attention from the language modeling community, continuous diffusion has appeared less scalable than discrete approaches. To challenge this belief we revisit Plaid, a likelihood-based...

<https://arxiv.org/abs/2605.18530>

15 CONF

Post-Trained MoE Can Skip Half Experts via Self-Distillation

ArXiv CS.AI 50%

arXiv:2605.18643v1 Announce Type: cross Abstract: Mixture-of-Experts (MoE) scales language models efficiently through sparse expert activation, and its dynamic variant further reduces computation by adjusting the activated experts in an input-dependent manner. Existing dynamic...

<https://arxiv.org/abs/2605.18643>

16 CONF

Lance: Unified Multimodal Modeling by Multi-Task Synergy

ArXiv CS.AI 50%

arXiv:2605.18678v1 Announce Type: cross Abstract: We present Lance, a lightweight native unified model supporting multimodal understanding, generation, and editing for both images and videos. Rather than relying on model capacity scaling or text-image-dominant designs, Lance...

<https://arxiv.org/abs/2605.18678>

17 CONF

Reversa: A Reverse Documentation Engineering Framework for Converting Legacy Software into Operational Specifications for AI Agents

ArXiv CS.AI 50%

arXiv:2605.18684v1 Announce Type: cross Abstract: Legacy systems concentrate business rules, architectural decisions, and operational exceptions that often remain implicit in code, data, configuration, and maintenance practices. At the same time, language-model-based coding...

<https://arxiv.org/abs/2605.18684>

18 CONF

PopPy: Opportunistically Exploiting Parallelism in Python Compound AI Applications

ArXiv CS.AI 50%

arXiv:2605.18697v1 Announce Type: cross Abstract: Compound AI applications, which compose calls to ML models using a general-purpose programming language like Python, are widely used for a variety of user-facing tasks, from software engineering to enterprise automation, making...

<https://arxiv.org/abs/2605.18697>

19 CONF

Predictable Confabulations: Factual Recall by LLMs Scales with Model Size and Topic Frequency

ArXiv CS.AI 50%

arXiv:2605.18732v1 Announce Type: cross Abstract: While scaling laws govern aggregate large language model performance, no scaling law has linked factual recall to both model size and training-data composition. We evaluated 38 models on over 8,900 scholarly references evaluated...

<https://arxiv.org/abs/2605.18732>

20 CONF

Principles of frugal inference and control

ArXiv CS.AI 50%

arXiv:2406.14427v4 Announce Type: replace Abstract: A central challenge for intelligent agents in an uncertain world is striking the right balance between utility maximization and resource use, not only for external movement but also for internal computation. Existing theories...

<https://arxiv.org/abs/2406.14427>

21 CONF

Evaluating AI Alignment in LLMs: Output Analysis of Value Priorities Across 75 Models with Human Benchmarking

ArXiv CS.AI 50%

arXiv:2506.12617v4 Announce Type: replace Abstract: Large language models (LLMs) are increasingly used in human-AI interaction research and practice, yet existing capability and safety benchmarks reveal little about the value priorities these systems express or how those...

<https://arxiv.org/abs/2506.12617>

22 CONF

OSWorld-Human: Benchmarking the Efficiency of Computer-Use Agents

ArXiv CS.AI 50%

arXiv:2506.16042v2 Announce Type: replace Abstract: Generative AI is being leveraged to solve a variety of computer-use tasks involving desktop applications. State-of-the-art systems have focused solely on improving accuracy on leading benchmarks. However, these systems are...

<https://arxiv.org/abs/2506.16042>

23 CONF

An AI system to help scientists write expert-level empirical software

ArXiv CS.AI 50%

arXiv:2509.06503v2 Announce Type: replace Abstract: The cycle of scientific discovery is frequently bottlenecked by the slow, manual creation of software to support computational experiments\cite{hannay2009how}. To address this, we present Empirical Research Assistance (ERA), an...

<https://arxiv.org/abs/2509.06503>

24 CONF

Position: AI Evaluations Should be Grounded on a Theory of Capability

ArXiv CS.AI 50%

arXiv:2509.19590v2 Announce Type: replace Abstract: Evaluations of generative models are now ubiquitous, and their outcomes critically shape public and scientific expectations of AI's capabilities. Yet skepticism about their reliability continues to grow. How can we know that a...

<https://arxiv.org/abs/2509.19590>

25 CONF

ECHO: Efficient Chest X-ray Report Generation with One-step Block Diffusion

ArXiv CS.AI 50%

arXiv:2604.09450v2 Announce Type: replace-cross Abstract: Chest X-ray report generation (CXR-RG) has the potential to substantially alleviate radiologists' workload. However, conventional autoregressive vision--language models (VLMs) suffer from high inference latency due to...

<https://arxiv.org/abs/2604.09450>

26 CONF

ALIGN: A Vision-Language Framework for High-Accuracy Accident Location Inference through Geo-Spatial Neural Reasoning

ArXiv CS.AI 50%

arXiv:2511.06316v3 Announce Type: replace Abstract: In low- and middle-income countries, public safety and urban planning initiatives frequently face a critical shortage of accurate, location-specific road crash data. Extracting reliable geospatial information from unstructured...

<https://arxiv.org/abs/2511.06316>

27 CONF

QuickLAP: Quick Language-Action Preference Learning for Semi-Autonomous Agents

ArXiv CS.AI 50%

arXiv:2511.17855v4 Announce Type: replace Abstract: Robots must learn from both what people do and what they say, but either modality alone is often incomplete: physical corrections are grounded but ambiguous in intent, while language expresses high-level goals but lacks...

<https://arxiv.org/abs/2511.17855>

28 CONF

What Drives Success in Physical Planning with Joint-Embedding Predictive World Models?

ArXiv CS.AI 50%

arXiv:2512.24497v3 Announce Type: replace Abstract: A long-standing challenge in AI is to develop agents capable of solving a wide range of physical tasks and generalizing to new, unseen tasks and environments. A popular recent approach involves training a world model from...

<https://arxiv.org/abs/2512.24497>

29 CONF

Med-V1: Small Language Models for Zero-shot and Scalable Biomedical Evidence Attribution

ArXiv CS.AI 50%

arXiv:2603.05308v2 Announce Type: replace-cross Abstract: Assessing whether an article supports an assertion is essential for hallucination detection and claim verification. While large language models (LLMs) have the potential to automate this task, achieving strong performance...

<https://arxiv.org/abs/2603.05308>

30 CONF

Agentic AI Governance and Lifecycle Management in Healthcare

ArXiv CS.AI 50%

arXiv:2601.15630v2 Announce Type: replace Abstract: Healthcare organizations are beginning to embed agentic AI into routine workflows, including clinical documentation support and early-warning monitoring. As these capabilities diffuse across departments and vendors, health...

<https://arxiv.org/abs/2601.15630>

31 CONF

Inference-Time Diversity in RL-Trained Lean Theorem Provers: A Diagnostic Study

ArXiv CS.AI 50%

arXiv:2601.16172v2 Announce Type: replace Abstract: RL-trained Lean theorem provers mode-collapse at inference time: on miniF2F-test with DeepSeek-Prover-V1.5-RL, doubling the i.i.d.\ sampling budget from $k=32$ to $k=64$ produces zero additional solved theorems (42/244 in...

<https://arxiv.org/abs/2601.16172>

32 CONF

The Alien Space of Science: Sampling Coherent but Cognitively Unavailable Research Directions

ArXiv CS.AI 50%

arXiv:2603.01092v2 Announce Type: replace Abstract: Scientific discovery is constrained not only by what is true, but by what is cognitively available to the researchers currently exploring a field. Many directions are coherent in light of the literature yet unlikely to be...

<https://arxiv.org/abs/2603.01092>

33 CONF

CodeScaler: Scaling Code LLM Training and Test-Time Inference via Reward Models

ArXiv CS.AI 50%

arXiv:2602.17684v2 Announce Type: replace-cross Abstract: Reinforcement Learning from Verifiable Rewards (RLVR) has driven recent progress in code large language models by leveraging execution-based feedback from unit tests, but its scalability is fundamentally constrained by...

<https://arxiv.org/abs/2602.17684>

34 CONF

A Comparative Study in Surgical AI: Potential and Limitations of Data, Compute, and Scaling

ArXiv CS.AI 50%

arXiv:2603.27341v3 Announce Type: replace Abstract: Recent Artificial Intelligence (AI) models have matched or exceeded human experts in several benchmarks of biomedical task performance, but surgical benchmarks in particular are often missing from prominent medical benchmark...

<https://arxiv.org/abs/2603.27341>

35 CONF

Breaking $\textit{Winner-Takes-All}$: Cooperative Policy Optimization Improves Diverse LLM Reasoning

ArXiv CS.AI 50%

arXiv:2605.11461v2 Announce Type: replace Abstract: Reinforcement learning with verifiers (RLVR) has become a central paradigm for improving LLM reasoning, yet popular group-based optimization algorithms like GRPO often suffer from exploration collapse, where the models...

<https://arxiv.org/abs/2605.11461>

36 CONF

AutoLLMResearch: Training Research Agents for Automating LLM Experiment Configuration - Learning from Cheap, Optimizing Expensive

ArXiv CS.AI 50%

arXiv:2605.11518v2 Announce Type: replace Abstract: Effectively configuring scalable large language model (LLM) experiments, spanning architecture design, hyperparameter tuning, and beyond, is crucial for advancing LLM research, as poor configuration choices can waste...

<https://arxiv.org/abs/2605.11518>

37 CONF

Model-Adaptive Tool Necessity Reveals the Knowing-Doing Gap in LLM Tool Use

ArXiv CS.AI 50%

arXiv:2605.14038v2 Announce Type: replace Abstract: Large language models (LLMs) increasingly act as autonomous agents that must decide when to answer directly vs. when to invoke external tools. Prior work studying adaptive tool use has largely treated tool necessity as a...

<https://arxiv.org/abs/2605.14038>

38 CONF

LightTransfer: Your Long-Context LLM is Secretly a Hybrid Model with Effortless Adaptation

ArXiv CS.AI 50%

arXiv:2410.13846v3 Announce Type: replace-cross Abstract: Scaling language models to handle longer contexts introduces substantial memory challenges due to the growing cost of key-value (KV) caches. Motivated by the efficiency gains of hybrid models and the broad availability of...

<https://arxiv.org/abs/2410.13846>

39 CONF

Uncertainty Quantification as a Principled Foundation for Explainable Artificial Intelligence: A Case Study of Counterfactual Explanations

ArXiv CS.AI 50%

arXiv:2502.17007v2 Announce Type: replace-cross Abstract: In this paper we argue that, to its detriment, transparency research overlooks many foundational concepts of artificial intelligence. As an illustrating example we focus on uncertainty quantification in the context of...

<https://arxiv.org/abs/2502.17007>

40 CONF

LLM-Safety Evaluations Lack Robustness

ArXiv CS.AI 50%

arXiv:2503.02574v2 Announce Type: replace-cross Abstract: In this paper, we argue that current safety alignment research efforts for large language models are hindered by many intertwined sources of noise, such as small datasets, methodological inconsistencies, and unreliable...

<https://arxiv.org/abs/2503.02574>

41 CONF

Patients Speak, AI Listens: LLM-based Analysis of Online Reviews Uncovers Key Drivers for Urgent Care Satisfaction

ArXiv CS.AI 50%

arXiv:2503.20981v2 Announce Type: replace-cross Abstract: Investigating the public experience of urgent care facilities is essential for promoting community healthcare development. Traditional survey methods often fall short due to limited scope, time, and spatial coverage....

<https://arxiv.org/abs/2503.20981>

42 CONF

Sustainability via LLM Right-sizing

ArXiv CS.AI 50%

arXiv:2504.13217v3 Announce Type: replace-cross Abstract: Large language models (LLMs) have become increasingly embedded in organizational workflows. This has raised concerns over their energy consumption, financial costs, and data sovereignty. While performance benchmarks often...

<https://arxiv.org/abs/2504.13217>

43 CONF

Catastrophic Overfitting, Entropy Gap and Participation Ratio: A Noiseless ℓ^p Norm Solution for Fast Adversarial Training

ArXiv CS.AI 50%

arXiv:2505.02360v2 Announce Type: replace-cross Abstract: Adversarial training is a cornerstone of robust deep learning, but fast methods like the Fast Gradient Sign Method (FGSM) often suffer from Catastrophic Overfitting (CO), where models become robust to single-step attacks...

<https://arxiv.org/abs/2505.02360>

44 CONF

HTSC-2025: A Benchmark Dataset of Ambient-Pressure High-Temperature Superconductors for AI-Driven Critical Temperature Prediction

ArXiv CS.AI 50%

arXiv:2506.03837v2 Announce Type: replace-cross Abstract: The discovery of high-temperature superconducting materials holds great significance for human industry and daily life. In recent years, research on predicting superconducting transition temperatures using artificial...

<https://arxiv.org/abs/2506.03837>

45 CONF

Algebraic Priors for Approximately Equivariant Networks

ArXiv CS.AI 50%

arXiv:2506.08244v2 Announce Type: replace-cross Abstract: Equivariant neural networks incorporate symmetries through group actions, embedding them as an inductive bias to improve performance. Existing methods learn an equivariant action on the latent space, or design...

<https://arxiv.org/abs/2506.08244>

46 CONF

Mixture-of-Experts Can Surpass Dense LLMs Under Strictly Equal Resource

ArXiv CS.AI 50%

arXiv:2506.12119v2 Announce Type: replace-cross Abstract: Mixture-of-Experts (MoE) language models dramatically expand model capacity and achieve remarkable performance without increasing per-token compute. However, can MoEs surpass dense architectures under strictly equal...

<https://arxiv.org/abs/2506.12119>

47 CONF

Missing-Modality-Aware Graph Neural Network for Cancer Classification

ArXiv CS.AI 50%

arXiv:2506.22901v2 Announce Type: replace-cross Abstract: A key challenge in learning from multimodal biological data is missing modalities, where data from one or more modalities are absent for some patients. Existing approaches either exclude patients with missing modalities,...

<https://arxiv.org/abs/2506.22901>

48 CONF

LSDTs: LLM-Augmented Semantic Digital Twins for Adaptive Knowledge-Intensive Infrastructure Planning

ArXiv CS.AI 50%

arXiv:2508.06799v3 Announce Type: replace-cross Abstract: Digital Twins (DTs) offer powerful tools for managing complex infrastructure systems, but their effectiveness is often limited by challenges in integrating unstructured knowledge. Recent advances in Large Language Models...

<https://arxiv.org/abs/2508.06799>

49 CONF

Lean Meets Theoretical Computer Science: Scalable Synthesis of Theorem Proving Challenges in Formal-Informal Pairs

ArXiv CS.AI 50%

arXiv:2508.15878v2 Announce Type: replace-cross Abstract: Formal theorem proving (FTP) has emerged as a critical foundation for evaluating the reasoning capabilities of large language models, enabling automated verification of mathematical proofs at scale. However, progress has...

<https://arxiv.org/abs/2508.15878>

50 CONF

OPERA: A Reinforcement Learning--Enhanced Orchestrated Planner-Executor Architecture for Reasoning-Oriented Multi-Hop Retrieval

ArXiv CS.AI 50%

arXiv:2508.16438v4 Announce Type: replace-cross Abstract: Recent advances in large language models (LLMs) and dense retrievers have driven significant progress in retrieval-augmented generation (RAG). However, existing approaches face significant challenges in complex...

<https://arxiv.org/abs/2508.16438>

51 CONF

Improving MLLM Training Efficiency via Stage-Aware Sparsity

ArXiv CS.AI 50%

arXiv:2509.18150v2 Announce Type: replace-cross Abstract: Multimodal Large Language Models (MLLMs) have demonstrated outstanding performance across a variety of domains. However, training MLLMs is often inefficient, as much of the computation is redundant due to the long input...

<https://arxiv.org/abs/2509.18150>

52 CONF

Can LLMs Refuse Questions They Do Not Know? Measuring Knowledge-Aware Refusal in Factual Tasks

ArXiv CS.AI 50%

arXiv:2510.01782v2 Announce Type: replace-cross Abstract: Large Language Models (LLMs) should refuse to answer questions beyond their knowledge. This capability, which we term knowledge-aware refusal, is crucial for factual reliability, while existing metrics fail to capture...

<https://arxiv.org/abs/2510.01782>

53 CONF

NeuroRVQ: Multi-Scale Biosignal Tokenization for Generative Foundation Models

ArXiv CS.AI 50%

arXiv:2510.13068v4 Announce Type: replace-cross Abstract: Biosignals such as electroencephalography (EEG), electrocardiography (ECG), and electromyography (EMG) encode physiological activity across multiple temporal and spectral scales, yielding representations that are rich but...

<https://arxiv.org/abs/2510.13068>

54 CONF

Prior Knowledge Makes It Possible: From Sublinear Graph Algorithms to LLM Test-Time Methods

ArXiv CS.AI 50%

arXiv:2510.16609v3 Announce Type: replace-cross Abstract: Test-time augmentation, such as Retrieval-Augmented Generation (RAG) or tool use, critically depends on an interplay between a model's parametric knowledge and externally retrieved information. However, the theoretical...

<https://arxiv.org/abs/2510.16609>

55 CONF

ProfBench: Multi-Domain Rubrics requiring Professional Knowledge to Answer and Judge

ArXiv CS.AI 50%

arXiv:2510.18941v2 Announce Type: replace-cross Abstract: Evaluating progress in large language models (LLMs) is often constrained by the challenge of verifying responses, limiting assessments to tasks like mathematics, programming, and short-form question-answering. However,...

<https://arxiv.org/abs/2510.18941>

56 CONF

Automated Coding of Communication Data Using ChatGPT: Consistency Across Subgroups

ArXiv CS.AI 50%

arXiv:2510.20584v3 Announce Type: replace-cross Abstract: Assessing communication and collaboration at scale depends on a labor-intensive task of coding communication data into categories according to different frameworks. Prior research has established that ChatGPT can be...

<https://arxiv.org/abs/2510.20584>

57 CONF

The Journal of Prompt-Engineered (Moral) Philosophy Or: Why AI-Assisted Ethics Research Requires Process Transparency

ArXiv CS.AI 50%

arXiv:2511.08639v3 Announce Type: replace-cross Abstract: Existing AI disclosure mandates in scholarship require that AI assistance be reported but leave transparency philosophically unspecified: they fix the duty without explaining what the duty serves. We argue that ethical...

<https://arxiv.org/abs/2511.08639>

58 CONF

GraphMind: Theorem Selection and Conclusion Generation Framework with Dynamic GNN for LLM Reasoning

ArXiv CS.AI 50%

arXiv:2511.19078v2 Announce Type: replace-cross Abstract: Large language models (LLMs) have demonstrated impressive capabilities in natural language understanding and generation, including multi-step reasoning such as mathematical proving. However, existing approaches often lack...

<https://arxiv.org/abs/2511.19078>

59 CONF

The Illusion of Specialization: Unveiling the Domain-Invariant "Standing Committee" in Mixture-of-Experts Models

ArXiv CS.AI 50%

arXiv:2601.03425v2 Announce Type: replace-cross Abstract: Mixture of Experts models are widely assumed to achieve domain specialization through sparse routing. In this work, we question this assumption by introducing COMMITTEEAUDIT, a post hoc framework that analyzes routing...

<https://arxiv.org/abs/2601.03425>

60 CONF

Watching, Reasoning, and Searching: A Video Deep Research Benchmark on Open Web for Agentic Video Reasoning

ArXiv CS.AI 50%

arXiv:2601.06943v2 Announce Type: replace-cross Abstract: In real-world video question answering scenarios, videos often provide only localized visual cues, while verifiable answers are distributed across the open web; models therefore need to jointly perform cross-frame clue...

<https://arxiv.org/abs/2601.06943>

61 CONF

Enhancing Cloud Network Resilience via a Robust LLM-Empowered Multi-Agent Reinforcement Learning Framework

ArXiv CS.AI 50%

arXiv:2601.07122v2 Announce Type: replace-cross Abstract: While virtualization and resource pooling empower cloud networks with structural flexibility and elastic scalability, they inevitably expand the attack surface and challenge cyber resilience. Reinforcement Learning...

<https://arxiv.org/abs/2601.07122>

62 CONF

Breaking the accuracy-resource dilemma: a lightweight adaptive video inference enhancement

ArXiv CS.AI 50%

arXiv:2601.14568v2 Announce Type: replace-cross Abstract: Existing video inference (VI) enhancement methods typically aim to improve performance by scaling up model sizes and employing sophisticated network architectures. While these approaches demonstrated state-of-the-art...

<https://arxiv.org/abs/2601.14568>

63 CONF

PyHealth 2.0: A Comprehensive Open-Source Toolkit for Accessible and Reproducible Clinical Deep Learning

ArXiv CS.AI 50%

arXiv:2601.16414v2 Announce Type: replace-cross Abstract: Difficulty replicating baselines, high computational costs, and required domain expertise create persistent barriers to clinical AI research. To address these challenges, we introduce PyHealth 2.0, an enhanced clinical...

<https://arxiv.org/abs/2601.16414>

64 CONF

Beyond Superficial Unlearning: Sharpness-Aware Robust Erasure of Hallucinations in Multimodal LLMs

ArXiv CS.AI 50%

arXiv:2601.16527v2 Announce Type: replace-cross Abstract: Multimodal LLMs are powerful but prone to object hallucinations, which describe non-existent entities and harm reliability. While recent unlearning methods attempt to mitigate this, we identify a critical flaw: structural...

<https://arxiv.org/abs/2601.16527>

65 CONF

On the Adversarial Robustness of Large Vision-Language Models under Visual Token Compression

ArXiv CS.AI 50%

arXiv:2601.21531v2 Announce Type: replace-cross Abstract: Visual token compression is widely used to accelerate large vision-language models (LVLMs) by pruning or merging visual tokens, yet its adversarial robustness remains unexplored. We show that existing encoder-based...

<https://arxiv.org/abs/2601.21531>

66 CONF

SuReNav: Superpixel Graph-based Constraint Relaxation for Navigation in Over-constrained Environments

ArXiv CS.AI 50%

arXiv:2602.06807v2 Announce Type: replace-cross Abstract: We address the over-constrained planning problem in semi-static environments. The planning objective is to find a best-effort solution that avoids all hard constraint regions while minimally traversing the least risky...

<https://arxiv.org/abs/2602.06807>

67 CONF

Trust the uncertain teacher: distilling dark knowledge via calibrated uncertainty

ArXiv CS.AI 50%

arXiv:2602.12687v2 Announce Type: replace-cross Abstract: The core of knowledge distillation lies in transferring the teacher's rich 'dark knowledge'-subtle probabilistic patterns that reveal how classes are related and the distribution of uncertainties. While this idea is well...

<https://arxiv.org/abs/2602.12687>

68 CONF

Self-Play Only Evolves When Self-Synthetic Pipeline Ensures Learnable Information Gain

ArXiv CS.AI 50%

arXiv:2603.02218v2 Announce Type: replace-cross Abstract: Large language models (LLMs) make it plausible to build systems that improve through self-evolving loops, but many existing proposals are better understood as self-play and often plateau quickly. A central failure mode is...

<https://arxiv.org/abs/2603.02218>

69 CONF

Old Habits Die Hard: How Conversational History Geometrically Traps LLMs

ArXiv CS.AI 50%

arXiv:2603.03308v2 Announce Type: replace-cross Abstract: How does the conversational past of large language models (LLMs) influence their future performance? Recent work suggests that LLMs are affected by their conversational history in unexpected ways. For instance,...

<https://arxiv.org/abs/2603.03308>

70 CONF

Are Multimodal LLMs Ready for Surveillance? A Reality Check on Zero-Shot Anomaly Detection in the Wild

ArXiv CS.AI 50%

arXiv:2603.04727v2 Announce Type: replace-cross Abstract: Multimodal large language models (MLLMs) have demonstrated impressive general competence in video understanding, yet their reliability for real-world Video Anomaly Detection (VAD) remains largely unexplored. Unlike...

<https://arxiv.org/abs/2603.04727>

71 CONF

The Expert Strikes Back: Interpreting Mixture-of-Experts Language Models at Expert Level

ArXiv CS.AI 50%

arXiv:2604.02178v2 Announce Type: replace-cross Abstract: Mixture-of-Experts (MoE) architectures have become the dominant choice for scaling Large Language Models (LLMs), activating only a subset of parameters per token. While MoE architectures are primarily adopted for...

<https://arxiv.org/abs/2604.02178>

72 CONF

Limitations of Sequence-Based Protein Representations for Parkinson's Disease Classification: A Leakage-Free Benchmark

ArXiv CS.AI 50%

arXiv:2604.11852v2 Announce Type: replace-cross Abstract: The identification of reliable molecular biomarkers for Parkinson's disease remains challenging due to its multifactorial nature. Although protein sequences constitute a fundamental and widely available source of...

<https://arxiv.org/abs/2604.11852>

73 CONF

CoLLM: Continuous Adaptation for SLO-Aware LLM Serving on Shared GPU Clusters

ArXiv CS.AI 50%

arXiv:2604.16400v2 Announce Type: replace-cross Abstract: As Large Language Models (LLMs) are increasingly adopted in edge intelligence to power domain-specific applications and personalized services, the quality and efficiency of the LLM post-training phase-including...

<https://arxiv.org/abs/2604.16400>

74 CONF

Beyond LoRA vs. Full Fine-Tuning: Gradient-Guided Optimizer Routing for LLM Adaptation

ArXiv CS.AI 50%

arXiv:2605.07111v2 Announce Type: replace-cross Abstract: Recent literature on fine-tuning Large Language Models highlights a fundamental debate. While Full Fine-Tuning (FFT) provides the representational plasticity required for high-entropy knowledge injection, Low-Rank...

<https://arxiv.org/abs/2605.07111>

75 CONF

MULTITEXTEDIT: Benchmarking Cross-Lingual Degradation in Text-in-Image Editing

ArXiv CS.AI 50%

arXiv:2605.08163v2 Announce Type: replace-cross Abstract: Text-in-image editing has become a key capability for visual content creation, yet existing benchmarks remain overwhelmingly English-centric and often conflate visual plausibility with semantic correctness. We introduce...

<https://arxiv.org/abs/2605.08163>

76 CONF

Missing Old Logits in Asynchronous Agentic RL: Semantic Mismatch and Repair Methods for Off-Policy Correction

ArXiv CS.AI 50%

arXiv:2605.12070v2 Announce Type: replace-cross Abstract: Asynchronous reinforcement learning improves rollout throughput for large language model agents by decoupling sample generation from policy optimization, but it also introduces a critical failure mode for PPO-style...

<https://arxiv.org/abs/2605.12070>

77 CONF

UAM: A Dual-Stream Perspective on Forgetting in VLA Training

ArXiv CS.AI 50%

arXiv:2605.15735v2 Announce Type: replace-cross Abstract: Vision--language--action (VLA) models are typically built by fine-tuning a pretrained vision--language model (VLM) on action data. However, we show that this standard recipe systematically erodes the VLM's multimodal...

<https://arxiv.org/abs/2605.15735>